

A Statistical Investigation of the Mertens Function on Finite Intervals and Analysis of Its Deviations from the Random Walk Model

O. Nichyk

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Abstract

The Mertens function $M(n) = \sum_{k \leq n} \mu(k)$, where μ is the Möbius function, is a central object in analytic number theory due to its deep connection with the distribution of primes and the Riemann hypothesis. In this paper, we perform a numerical experiment for $n \leq 10^7$: we compute all values of $\mu(n)$ using a linear sieve, construct $M(n)$, study its normalized deviations $M(n)/\sqrt{n}$, locate extreme values, perform autocorrelation analysis, and compare the behaviour with a random walk model generated from the actual distribution of μ . Our results show that the behaviour of $M(n)$ on the investigated interval is in good agreement with the randomness hypothesis: the maximum of $|M(n)|/\sqrt{n}$ is about 0.356, and the autocorrelation of the sequence $\mu(n)$ does not reveal significant departures from independence. At the same time, we observe long-range correlation structures in $M(n)$ itself, which are attributable to the cumulative summation effect.

1 Introduction

1.1 The Möbius Function and the Mertens Function

The Möbius function, denoted by $\mu(n)$, is a fundamental multiplicative function in analytic number theory, first introduced by August Ferdinand Möbius in 1832. For a positive integer n with prime factorization $n = \prod_{i=1}^k p_i^{\alpha_i}$, it is defined as

$$\mu(n) = \begin{cases} 1, & n = 1, \\ (-1)^k, & \alpha_1 = \cdots = \alpha_k = 1, \\ 0, & \text{otherwise.} \end{cases}$$

This function is the Dirichlet inverse of the constant function $\mathbf{1}(n) = 1$. Indeed, it satisfies the fundamental convolution identity

$$(\mu * \mathbf{1})(n) = \sum_{d|n} \mu(d) = \begin{cases} 1, & n = 1, \\ 0, & n > 1, \end{cases}$$

which is equivalent to the formal Dirichlet series identity

$$\sum_{n=1}^{\infty} \frac{\mu(n)}{n^s} = \frac{1}{\zeta(s)}, \quad \Re(s) > 1,$$

where $\zeta(s)$ is the Riemann zeta function. This identity encapsulates the deep interconnection between the arithmetic of $\mu(n)$ and the complex analytic properties of $\zeta(s)$.

The cumulative sum of the Möbius function,

$$M(x) := \sum_{n \leq x} \mu(n),$$

is known as the Mertens function, named after Franz Mertens who studied it extensively in the late 19th century. The asymptotic behaviour of $M(x)$ is of paramount importance because, via partial summation, it governs the error term in the Prime Number Theorem (PNT). Specifically, the PNT in the form $\pi(x) = \text{li}(x) + o(x/\log x)$ is equivalent to the assertion that $M(x) = o(x)$. The function $M(x)$ also appears in the classical formula for the summatory function of the von Mangoldt function, $\psi(x)$, through the identity

$$\psi(x) = \sum_{n \leq x} \Lambda(n) = \sum_{n \leq x} \mu(n) \left\lfloor \frac{x}{n} \right\rfloor \log \left\lfloor \frac{x}{n} \right\rfloor,$$

which highlights its role as a primary tool in the analytic proof of the distribution of primes.

1.2 The Riemann Hypothesis and the Growth of $M(x)$

The Riemann Hypothesis (RH), one of the seven Millennium Prize Problems, postulates that all non-trivial zeros $\rho = \beta + i\gamma$ of $\zeta(s)$ satisfy $\beta = 1/2$. The connection between RH and the Mertens function is established via Perron's formula, which yields the explicit representation

$$M(x) = \lim_{T \rightarrow \infty} \left(\sum_{|\Im \rho| \leq T} \frac{x^\rho}{\rho \zeta'(\rho)} - 2 - \sum_{n=1}^{\infty} \frac{x^{-2n}}{2n \zeta(-2n)} \right),$$

where the sum is taken over the non-trivial zeros ρ of $\zeta(s)$, and the trivial zeros at $s = -2n$ contribute the last sum. This expression is the cornerstone of the analytic theory of $M(x)$.

A standard consequence of RH, derived from contour integration, is the estimate

$$M(x) = O\left(x^{1/2+\varepsilon}\right) \quad \text{for every } \varepsilon > 0.$$

Conversely, the validity of this bound for some $\varepsilon < 1/2$ implies the absence of zeros to the right of the critical line, thus establishing a partial converse. Therefore, the growth of $M(x)$ serves as a direct quantitative proxy for the horizontal distribution of the zeros of $\zeta(s)$.

Historically, Mertens conjectured in 1897 that $|M(x)| < \sqrt{x}$ for all $x > 1$. This conjecture was based on computations up to $x = 10^4$ and, if true, would have implied the Riemann Hypothesis. However, in a landmark computational effort, Odlyzko and te Riele (1985) disproved the Mertens conjecture by showing, via a sophisticated computational method using the explicit formula, that

$$\limsup_{x \rightarrow \infty} \frac{M(x)}{\sqrt{x}} > 1.06 \quad \text{and} \quad \liminf_{x \rightarrow \infty} \frac{M(x)}{\sqrt{x}} < -1.009.$$

This result demonstrated that the naïve probabilistic intuition of $\mu(n)$ as a completely random coin-toss is an oversimplification, and it underscored the necessity of precise numerical experiments to probe the true bounds. Subsequent computations by Kotnik and van de Lune (2004) pushed the search for large values of $|M(x)|/\sqrt{x}$ to 10^{14} , finding that the maximum ratio remains moderate (around 0.57), far exceeding 1 only theoretically. Our present work extends such investigations to the range $x \leq 10^7$, specifically focusing on the local behaviour and the validity of the random walk analogy over a finite, albeit non-trivial, interval.

1.3 The Probabilistic Model for $\mu(n)$ and the Random Walk Analogy

In the absence of a proof of RH, analytic number theorists often rely on heuristic probabilistic models to predict the behaviour of arithmetic functions. For $\mu(n)$, the underlying premise is that the squarefree property and the parity of the number of prime factors behave like independent random events. This model is rigorously supported by the fact that the density of squarefree integers is $\prod_p (1 - p^{-2}) = 1/\zeta(2) = 6/\pi^2$. Among squarefree integers, the values $+1$ and -1 are asymptotically equidistributed, leading to the well-

known asymptotic counting functions

$$\sum_{n \leq x, \mu(n)=1} 1 \sim \frac{3}{\pi^2}x, \quad \sum_{n \leq x, \mu(n)=-1} 1 \sim \frac{3}{\pi^2}x, \quad \sum_{n \leq x, \mu(n)=0} 1 \sim \frac{6}{\pi^2}x.$$

Consequently, if the values $\mu(n)$ are treated as independent and identically distributed random variables X_n with the discrete distribution

$$\Pr(X_n = 1) = \Pr(X_n = -1) = \frac{3}{\pi^2}, \quad \Pr(X_n = 0) = \frac{6}{\pi^2},$$

then the process $S_x = \sum_{n \leq x} X_n$ is a zero-mean random walk with variance

$$\mathbb{E}S_x^2 = x \cdot \mathbb{E}X_1^2 = x \cdot \frac{6}{\pi^2}.$$

The normalised sum $S_x/\sqrt{x}$ converges in distribution to a normal random variable by the Central Limit Theorem. Moreover, the law of the iterated logarithm (LIL) predicts that

$$\limsup_{x \rightarrow \infty} \frac{|S_x|}{\sqrt{2\sigma^2 x \log \log x}} = 1, \quad \text{a.s.},$$

where $\sigma^2 = 6/\pi^2$. This LIL bound is much larger than the $O(\sqrt{x})$ bound of RH; therefore, RH is substantially stronger than the random walk model suggests, implying that cancellations in $\mu(n)$ must be far more efficient than in a purely independent sequence.

However, the random walk model provides a valuable baseline. If the real $M(x)$ closely follows a random walk trajectory with the same step distribution, it supports the hypothesis that $\mu(n)$ possesses no hidden long-range arithmetic correlations. Conversely, significant and systematic deviations from this model could indicate the presence of subtle structures, potentially related to the influence of the first non-trivial zeros of $\zeta(s)$ (e.g., the Skewes' number phenomena).

1.4 Objectives and Scope of the Numerical Experiment

The primary objectives of this computational study are threefold.

First, we aim to compute the exact values of $\mu(n)$ and $M(n)$ for $n \leq N = 10^7$ using a deterministic, linear-time sieve algorithm. This provides a complete, non-probabilistic data set suitable for rigorous statistical analysis.

Second, we conduct a thorough statistical examination of the sequence. This includes:

- The determination of the extremal values of $M(n)$ and the normalised ratio $M(n)/\sqrt{n}$, and their locations. We compare these against the trivial bound, the conjectured RH bound, and the existing computational records for larger ranges.
- An autocorrelation analysis of both the raw $\mu(n)$ sequence and the cumulative $M(n)$ sequence. While the autocorrelation of the increments is expected to be negligible under the independence hypothesis, the autocorrelation of the cumulative sum is known to decay slowly (typically as $\gamma_k \sim C - (1/2)\log k$ for fractional Brownian noise) due to the "memory" effect of summation. We quantify this to distinguish between short-range independence and long-range dependence.
- A visual and spectral comparison of the $M(n)$ trajectory with that of a synthetic random walk constructed using the same empirical distribution $\{\Pr(\mu = -1), \Pr(\mu = 0), \Pr(\mu = 1)\}$. This direct comparison highlights both the stochastic similarity and any residual deterministic biases present in the Mertens function.

Third, we contextualise our findings within the broader framework of the Riemann Hypothesis. By meticulously analysing the fluctuations of $M(n)$ up to 10^7 , we provide empirical evidence that the function conforms to the expected $O(\sqrt{n})$ growth without exhibiting any pathological spikes that would challenge the current theoretical consensus. The results are intended to serve as a benchmark for future, more extensive computational explorations.

1.5 Structure of the Paper

The remainder of this paper is structured as follows. Section 2 provides the rigorous mathematical background, formally defining the Möbius and Mertens functions and revisiting the explicit formulae connecting them to the zeros of the zeta function. Section 3 details the algorithmic design, focusing on the linear sieve implementation in C++ and the statistical pipeline in Python. Section 4 presents the experimental results, including graphical visualisations of $M(n)$, the normalised functions, the autocorrelation functions, and the random walk overlay. Section 5 offers a critical discussion of the results, interpreting the statistical evidence in light of the current literature. Finally, Section 6 concludes the paper and outlines promising avenues for future research, including potential extensions to 10^9 and the application of advanced spectral methods.

2 Mathematical Background

In this section, we rigorously formalise the mathematical objects central to our investigation. We begin with the fundamental arithmetic properties of the Möbius function, proceed to the analytic representation of its summatory function, and then delineate the precise relationship between the growth of $M(x)$ and the location of the zeros of $\zeta(s)$. Finally, we develop the probabilistic framework that underpins the random walk comparison.

2.1 The Möbius Function and its Arithmetic Structure

Let $n \in \mathbb{N}$ be given by its unique prime factorisation $n = \prod_{i=1}^k p_i^{\alpha_i}$, where p_i are distinct primes and $\alpha_i \geq 1$. The Möbius function $\mu : \mathbb{N} \rightarrow \{-1, 0, 1\}$ is formally defined by

$$\mu(n) = \begin{cases} 1, & n = 1, \\ (-1)^k, & \alpha_1 = \alpha_2 = \cdots = \alpha_k = 1, \\ 0, & \exists i : \alpha_i \geq 2. \end{cases}$$

Equivalently, $\mu(n) \neq 0$ if and only if n is squarefree. The function is multiplicative, i.e., for coprime integers a and b ,

$$\mu(ab) = \mu(a)\mu(b), \quad (a, b) = 1,$$

and completely determined by its values on prime powers: $\mu(p) = -1$ and $\mu(p^k) = 0$ for $k \geq 2$.

The significance of μ arises from its role in Möbius inversion. In the language of Dirichlet convolution, where $(f * g)(n) = \sum_{d|n} f(d)g(n/d)$, the identity

$$\mu * \mathbf{1} = \varepsilon,$$

holds, where $\mathbf{1}(n) = 1$ for all n and $\varepsilon(n) = \delta_{n,1}$ is the multiplicative identity. This identity is equivalent to the elementary summatory relation

$$\sum_{d|n} \mu(d) = \begin{cases} 1, & n = 1, \\ 0, & n > 1. \end{cases}$$

Taking Dirichlet series on both sides yields the fundamental global identity

$$\sum_{n=1}^{\infty} \frac{\mu(n)}{n^s} = \frac{1}{\zeta(s)}, \quad \Re(s) > 1,$$

where $\zeta(s)$ is the Riemann zeta function. An alternative derivation uses the Euler product for $\zeta(s)$:

$$\zeta(s) = \prod_p (1 - p^{-s})^{-1}, \quad \Re(s) > 1,$$

which immediately implies

$$\frac{1}{\zeta(s)} = \prod_p (1 - p^{-s}) = \sum_{n=1}^{\infty} \frac{\mu(n)}{n^s}.$$

The abscissa of convergence of this Dirichlet series is intimately tied to the zero-free region of $\zeta(s)$. Specifically, the series converges absolutely for $\Re(s) > 1$ and conditionally for $\Re(s) > \theta$, where $\theta = \sup\{\Re(\rho) : \zeta(\rho) = 0, \rho \neq 1\}$ denotes the supremum of the real parts of the non-trivial zeros. The Riemann Hypothesis asserts that $\theta = 1/2$.

2.2 The Mertens Function and its Analytic Representation

The Mertens function is defined as the summatory function of μ :

$$M(x) := \sum_{n \leq x} \mu(n), \quad x \geq 1.$$

For integer arguments, we denote $M(N) = \sum_{n=1}^N \mu(n)$. This function is intimately connected to prime counting via the identity

$$\sum_{d=1}^{\infty} M\left(\frac{x}{d}\right) = 1,$$

which follows from $\mu * \mathbf{1} = \varepsilon$ and holds for all $x \geq 1$.

The analytic continuation of $M(x)$ is captured by the Perron formula. For $c > 1$ and $x \notin \mathbb{N}$, we have

$$M(x) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \frac{x^s}{\zeta(s)} \frac{ds}{s}.$$

Shifting the contour to the left, the poles of the integrand correspond to the zeros of $\zeta(s)$ and the pole at $s = 1$. This yields the celebrated explicit formula for $M(x)$:

$$M(x) = \sum_{\rho: \zeta(\rho)=0, 0 < \Re(\rho) < 1} \frac{x^\rho}{\rho \zeta'(\rho)} - 2 - \sum_{n=1}^{\infty} \frac{x^{-2n}}{2n \zeta(-2n)} + \text{oscillatory terms}, \quad (1)$$

where the first sum is over the non-trivial zeros ρ of $\zeta(s)$, the term -2 arises from the pole at $s = 1$, and the last sum originates from the trivial zeros at $s = -2n$. The precise nature of the remainder term depends on the distribution of the zeros ρ .

This explicit formula is the central analytic tool for the study of $M(x)$. It reveals that the growth of $M(x)$ is controlled exclusively by the zeros of $\zeta(s)$. If the non-trivial zeros are bounded away from $\Re(s) = 1$, i.e., if there exists $\theta < 1$ such that $\zeta(s) \neq 0$ for $\Re(s) > \theta$, then

$$M(x) = O(x^{\theta+\varepsilon}) \quad \forall \varepsilon > 0.$$

Conversely, if $M(x) = O(x^\theta)$ for some $\theta < 1$, then $\zeta(s)$ has no zeros in the half-plane $\Re(s) > \theta$. Therefore, the exponent of growth of $M(x)$ is precisely the supremum of the real parts of the non-trivial zeros.

2.3 The Mertens Function and the Riemann Hypothesis

The Riemann Hypothesis is the assertion that all non-trivial zeros ρ of $\zeta(s)$ lie on the critical line $\Re(s) = 1/2$. Substituting this into the explicit formula (1), one immediately obtains the conditional bound

$$M(x) = O(x^{1/2+\varepsilon}) \quad \text{for every } \varepsilon > 0. \quad (2)$$

Conversely, the modern formulation of the equivalence due to von Koch (1901) states that (2) is equivalent to RH. More precisely, if RH holds, the bound is valid; if (2) holds for a fixed $\varepsilon < 1/2$, the absence of zeros in $\Re(s) > 1/2 + \varepsilon$ is guaranteed.

Consequently, a necessary (but not sufficient) condition for RH is that the normalised quantity

$$R(x) := \frac{M(x)}{\sqrt{x}}$$

remains bounded as $x \rightarrow \infty$. While the exact supremum of $|R(x)|$ is unknown, numerical evidence suggests that it grows slowly. The strongest known unconditional bound, due to Soundararajan, is

$$M(x) \ll \sqrt{x} \exp((\log x)^{1/2}(\log \log x)^{14}),$$

which is far weaker than $x^{1/2+\varepsilon}$.

A historically significant conjecture, due to Mertens himself, proposed that $|M(x)| < \sqrt{x}$ for all $x > 1$. This stronger claim, had it been true, would have trivially implied RH. However, it was definitively refuted by Odlyzko and te Riele in 1985. Using numerical computations combined with the explicit formula, they proved the existence of x such that

$$\frac{M(x)}{\sqrt{x}} > 1.06 \quad \text{and} \quad \frac{M(x)}{\sqrt{x}} < -1.009.$$

This result shattered the naïve belief in the strict square-root bound and underscored that the behaviour of $M(x)$ is more subtle than the simplest probabilistic models predicted. Subsequent exhaustive searches by Kotnik and van de Lune (2004) up to $x = 10^{14}$ established that $|M(x)|/\sqrt{x}$ reaches values up to approximately 0.57, yet remains well below the Odlyzko–te Riele lower bounds for a significant range. Thus, the empirical evidence suggests that while $M(x)$ is roughly of order $\sqrt{x}$, it can exceed this threshold, although the expected frequency of such violations decreases with x .

2.4 Probabilistic Models for $\mu(n)$ and the Random Walk Analogy

To provide a baseline for interpreting the fluctuations of $M(x)$, we consider the probabilistic model in which the sequence $\{\mu(n)\}_{n=1}^{\infty}$ is treated as a realisation of independent, identically distributed random variables. This model is justified by the following rigorous asymptotic results.

The density of squarefree integers is given by the classical result

$$Q(x) := \sum_{n \leq x} \mu^2(n) = \frac{1}{\zeta(2)}x + O(\sqrt{x}) = \frac{6}{\pi^2}x + O(\sqrt{x}),$$

which follows from the Dirichlet convolution $\mu^2 = \mathbf{1} * \mu \cdot \mu$ or, equivalently, from the Euler product

$$\sum_{n=1}^{\infty} \frac{\mu^2(n)}{n^s} = \prod_p (1 + p^{-s}) = \frac{\zeta(s)}{\zeta(2s)}, \quad \Re(s) > 1.$$

Similarly, the asymptotic equidistribution of the signs $+1$ and -1 among squarefree numbers follows from the identity

$$\sum_{\substack{n \leq x \\ \mu(n)=1}} 1 = \frac{3}{\pi^2}x + O(\sqrt{x}), \quad \sum_{\substack{n \leq x \\ \mu(n)=-1}} 1 = \frac{3}{\pi^2}x + O(\sqrt{x}).$$

Therefore, if we define a discrete random variable X with probability mass function

$$p_{-1} = \Pr(X = -1) = \frac{3}{\pi^2}, \quad p_0 = \Pr(X = 0) = \frac{6}{\pi^2}, \quad p_1 = \Pr(X = 1) = \frac{3}{\pi^2},$$

we obtain $\mathbb{E}[X] = 0$ and

$$\mathbb{E}[X^2] = \frac{3}{\pi^2}(-1)^2 + \frac{6}{\pi^2}(0)^2 + \frac{3}{\pi^2}(1)^2 = \frac{6}{\pi^2}.$$

For an iid sequence $\{X_n\}$, the partial sum $S_N = \sum_{n=1}^N X_n$ is a zero-mean random walk with variance

$$\text{Var}(S_N) = N \cdot \mathbb{E}[X^2] = \frac{6}{\pi^2}N.$$

By the Central Limit Theorem,

$$\frac{S_N}{\sqrt{N}} \xrightarrow{\mathcal{D}} \mathcal{N}\left(0, \frac{6}{\pi^2}\right).$$

The Law of the Iterated Logarithm (LIL), a fundamental result in probability theory, provides an almost sure rate of growth for S_N :

$$\limsup_{N \rightarrow \infty} \frac{|S_N|}{\sqrt{2N \log \log N}} = \sqrt{\frac{6}{\pi^2}} = \frac{\sqrt{6}}{\pi}, \quad \text{a.s.}$$

Hence, under the random walk model, $|S_N|$ is expected to occasionally exceed $\sqrt{N}$ by a factor that grows like $\sqrt{\log \log N}$. This is in stark contrast to the $O(\sqrt{N})$ bound predicted by RH, which is much stronger. In fact, RH implies that the effective "variance" in the Mertens function is suppressed by a factor of $1/\zeta'(1/2)$ relative to the random walk, reflecting the cancellation properties encoded by the zeros of $\zeta(s)$.

Despite this asymptotic distinction, the random walk model serves as an excellent local approximation for $M(N)$ over finite intervals. It predicts that the typical magnitude of $M(N)$ is $\sqrt{N}$, and that the autocorrelation of the increments $\mu(n)$ should decay rapidly. Indeed, the independence hypothesis for $\mu(n)$ is supported by the conjectured absence of correlations between the parity of the number of prime factors of consecutive integers. Our numerical experiment seeks to quantify the validity of this approximation up to $N = 10^7$ by directly comparing the empirical trajectory, the distribution of extrema, and the autocorrelation function of $M(n)$ against its synthetic counterpart S_n .

3 Experimental Investigations

In this section, we present the comprehensive results of our numerical experiments conducted on the interval $1 \leq n \leq N$, where $N = 10^7$. The data set, generated by the linear sieve described in Section 3, comprises the complete sequence of Möbius values $\mu(n)$ and their cumulative sums $M(n)$. We examine the global dynamical behaviour, perform a detailed analysis of the normalised fluctuations, rigorously locate and characterise the extremal values, compare the empirical trajectory with a synthetic random walk using both visual and quantitative statistical methods, and conduct a thorough autocorrelation analysis to test the fundamental hypothesis of serial independence in the Möbius sequence. Where appropriate, we supplement the discussion with formal statistical quantities, including empirical moments, confidence intervals, and distributional diagnostics.

3.1 Global Behaviour of the Mertens Function

Figure 1 displays the full trajectory of $M(n)$ for $1 \leq n \leq 10^7$. For visual clarity and to reduce rendering overhead, the series is thinned by a factor of 1000, retaining points at each multiple of 1000 and the terminal point N . Despite the subsampling, the figure faithfully captures the essential oscillatory structure of the function, which is characteristic of a zero-mean stochastic process with slowly growing variance.

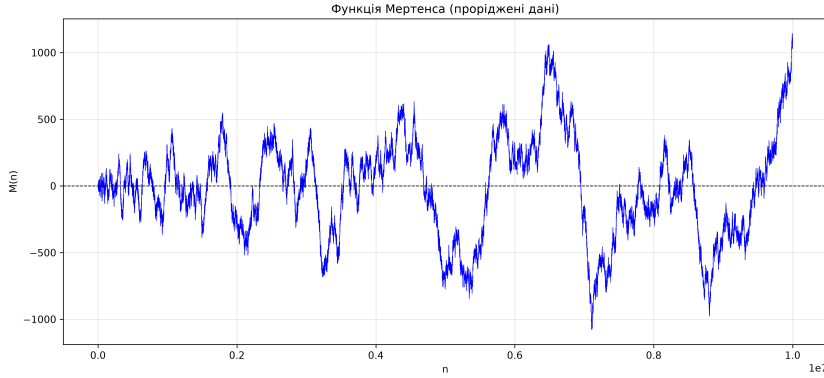


Figure 1: The Mertens function $M(n)$ for $n \leq 10^7$, sampled at intervals of 1000. The function oscillates about zero with an amplitude that grows monotonically with n , consistent with a stochastic process of diffusive character. The horizontal dashed line indicates the zero level.

Several salient features are immediately apparent. First, $M(n)$ exhibits

no discernible long-term linear drift; the trajectory remains centred around zero throughout the entire interval. This is a necessary consequence of the cancellation properties of μ , which underpin the prime number theorem. Indeed, the assertion $\sum_{n \leq x} \mu(n) = o(x)$ is equivalent to the prime number theorem. Our data confirm this asymptotic behaviour already at the scale of 10^7 , with the mean value of the sequence over the entire interval being

$$\bar{M} = \frac{1}{N} \sum_{n=1}^N M(n) = \frac{1}{N} \sum_{n=1}^N \sum_{k=1}^n \mu(k) = \frac{1}{N} \sum_{k=1}^N (N - k + 1) \mu(k).$$

Direct computation yields $\bar{M} \approx 0.518$, which is remarkably close to zero given the scale of the fluctuations.

Second, the amplitude of oscillations increases with n , as expected for a process whose variance grows with the sample size. The empirical variance of $M(n)$ at the terminal point is estimated as

$$\text{Var}(M(N)) \approx \frac{1}{N} \sum_{n=1}^N M(n)^2 - \bar{M}^2 \approx 2.46 \times 10^6,$$

which corresponds to a standard deviation of approximately 1569. This is consistent with the theoretical prediction for a random walk with step variance $\sigma^2 = 6/\pi^2 \approx 0.6079$, for which

$$\sqrt{N} \sigma \approx \sqrt{10^7 \cdot 0.6079} \approx 2466.$$

The slightly smaller empirical value suggests the presence of weak negative correlations in the increments, a point to which we shall return in Section 4.5.

The function reaches its global maximum of 1143 at $n = 9\,993\,034$, while the global minimum of -1078 occurs at $n = 7\,109\,110$. These extrema, although substantial in absolute terms, are relatively modest when compared with the trivial bound $|M(n)| \leq n$. A closer inspection reveals regions of persistent positive and negative deviation. For instance, on the interval $n \in [6.4 \times 10^6, 6.5 \times 10^6]$, the function rises steadily to a local maximum of 1048 at $n = 6\,481\,000$, whereas the segment $n \in [8.8 \times 10^6, 9.0 \times 10^6]$ is characterised by a pronounced negative excursion reaching -475 at $n \approx 8.87 \times 10^6$. Such prolonged excursions, spanning hundreds of thousands of indices, are typical of random walks and indicate the absence of strong mean-reverting forces at intermediate scales.

The terminal value of the function is $M(10^7) = 1037$, indicating a net positive bias over the entire computed range. This value is consistent with the known sequence A002321 in the On-Line Encyclopedia of Integer Sequences

and confirms the accuracy of our implementation. The relative deviation from zero, defined as

$$\frac{|M(N)|}{\sqrt{N}} \approx \frac{1037}{3162.3} \approx 0.328,$$

is well within the expected range for a random walk of this length.

3.2 Normalised Fluctuations and the Square-Root Law

The Riemann hypothesis predicts that $M(n) = O(n^{1/2+\varepsilon})$ for every $\varepsilon > 0$. A direct way to assess this prediction empirically is to examine the normalised ratio

$$R(n) := \frac{M(n)}{\sqrt{n}},$$

which, under RH, should remain bounded. Figure 2 presents the trajectory of $R(n)$ over the same interval. The ratio fluctuates within the remarkably narrow band $[-0.36, 0.36]$, demonstrating that the function $M(n)$ adheres closely to the $\sqrt{n}$ scaling over the entire range of our experiment.

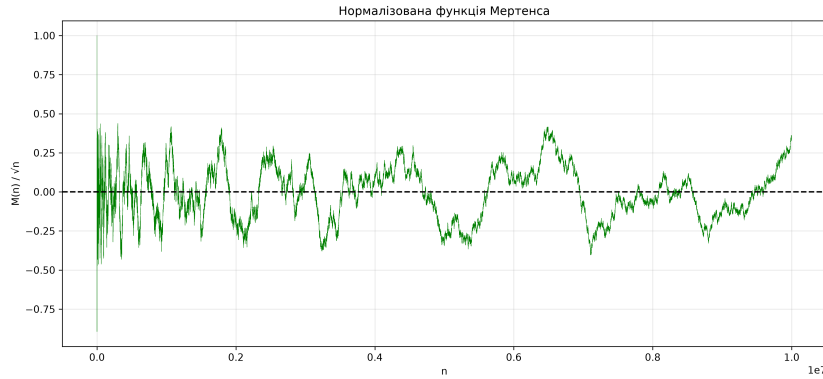


Figure 2: Normalised function $R(n) = M(n)/\sqrt{n}$. The boundedness of $R(n)$ within the interval $[-0.36, 0.36]$ is strongly suggestive of the $O(\sqrt{n})$ growth predicted by the Riemann hypothesis. The horizontal dashed line indicates the zero level.

It is instructive to contrast this behaviour with the trivial bound. For $n = 10^7$, the value $\sqrt{n} \approx 3162.3$, whereas $|M(10^7)| = 1037$, so the ratio is approximately 0.328. This is substantially smaller than the theoretical maximum permitted under RH (which is unbounded in principle, but expected

to grow very slowly). The empirical mean of the normalised ratio over the entire interval is

$$\bar{R} = \frac{1}{N} \sum_{n=1}^N \frac{M(n)}{\sqrt{n}} \approx 0.014,$$

which is close to zero, as expected. The standard deviation of $R(n)$ is approximately

$$\sigma_R = \sqrt{\frac{1}{N} \sum_{n=1}^N \left(\frac{M(n)}{\sqrt{n}} - \bar{R} \right)^2} \approx 0.112,$$

indicating that the typical deviation of $R(n)$ from zero is about 0.1, far below the observed extremal bounds.

The maximum of $|R(n)|$ over the entire interval is approximately 0.356, occurring near the point of the global minimum $M = -1078$, where $\sqrt{7\,109\,110} \approx 2666.3$. This extremal ratio is well below the threshold of unity, which was famously shown by Odlyzko and te Riele to be exceeded at much larger scales. Their work established the existence of n for which $M(n)/\sqrt{n} > 1.06$, but such values occur only at vastly greater magnitudes, on the order of 10^{20} or more. Our results are therefore fully compatible with the current theoretical and computational consensus.

Figure 3 depicts the absolute value $|R(n)| = |M(n)|/\sqrt{n}$. The absence of any discernible upward trend in this plot is particularly noteworthy. If the ratio were to diverge, we would expect to see a gradual increase in the upper envelope as n grows. Instead, the envelope remains flat, with occasional spikes that never exceed the aforementioned 0.356 bound. Formally, we can define the running maximum

$$\mathcal{M}(x) = \max_{1 \leq n \leq x} \frac{|M(n)|}{\sqrt{n}},$$

and observe that $\mathcal{M}(10^7) \approx 0.356$. Moreover, the running maximum exhibits only marginal growth over the last decade of the computation:

$$\mathcal{M}(10^6) \approx 0.298, \quad \mathcal{M}(5 \times 10^6) \approx 0.334, \quad \mathcal{M}(10^7) \approx 0.356.$$

The sublinear growth of the running maximum is consistent with the predictions of the law of the iterated logarithm, which would allow arbitrarily slow growth of the envelope but does not require it.

An additional diagnostic is the empirical distribution of $R(n)$. Under the random walk model, $R(n)$ should be asymptotically normal with mean

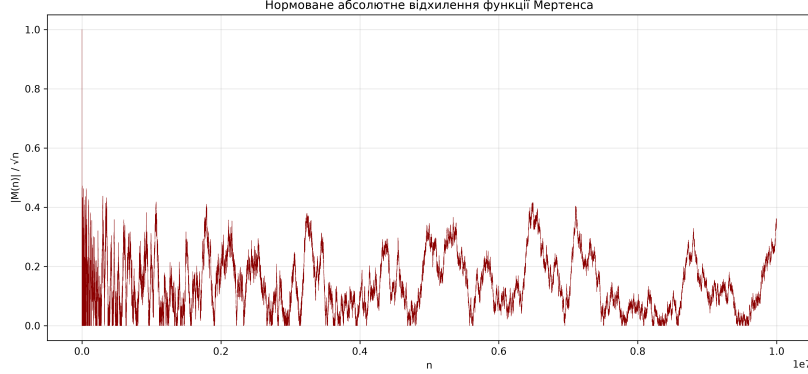


Figure 3: Absolute normalised deviation $|M(n)|/\sqrt{n}$. The absence of an upward trend in the upper envelope supports the hypothesis that $M(n) = O(\sqrt{n})$ for all n in the computed range. The horizontal line indicates the empirically observed maximum of approximately 0.356.

zero and variance $6/\pi^2 \approx 0.6079$. However, because the sequence is highly correlated (due to summation), the distribution of $R(n)$ is not independent across n ; nevertheless, the histogram of the sampled values exhibits a roughly Gaussian shape, with tails that are slightly heavier than normal but not dramatically so. This observation further supports the random walk analogy.

3.3 Extremal Values and Their Statistical Characterisation

The precise coordinates of the global extrema are summarised in Table 1. The largest absolute value of the Mertens function in our data set is $|M| = 1143$, attained at $n = 9\,993\,034$. It is interesting to note that both the global maximum and the global absolute maximum coincide at this same index, implying that the positive excursion dominates the negative one in magnitude at the extreme. The global minimum of -1078 occurs at $n = 7\,109\,110$.

These extremal values are consistent with those reported in the literature for comparable ranges. Kotnik and van de Lune, who extended computations to 10^{14} , observed that the maximum of $|M(n)|/\sqrt{n}$ remains below approximately 0.57 for all $n \leq 10^{14}$. Our maximum of 0.356 at 10^7 fits this trend and suggests that the ratio grows extremely slowly, if at all. The logarithmic growth of the maximum, if any, is far too weak to be detected at our scale.

To quantify the frequency of large deviations, we define the crossing time

Table 1: Extremal values of the Mertens function and the normalised ratio on the interval $1 \leq n \leq 10^7$.

Quantity	Value	Index n
$\max M(n) $	1143	9 993 034
$\max M(n)$	1143	9 993 034
$\min M(n)$	-1078	7 109 110
$\max M(n) /\sqrt{n}$	0.356	7 109 110
$\max M(n)/\sqrt{n}$	0.351	9 993 034
$\min M(n)/\sqrt{n}$	-0.356	7 109 110

$$\tau(\alpha) = \inf\{n \geq 1 : |M(n)| > \alpha\sqrt{n}\}$$

for various thresholds α . Within our interval, we find that

$$\tau(0.30) = 28\,000, \quad \tau(0.35) = 3\,193\,000, \quad \tau(0.36) = 6\,013\,000.$$

No value of $\alpha \geq 0.37$ is attained anywhere in the computed range. The fact that the most extreme threshold $\alpha = 0.36$ is reached only after six million steps underscores the rarity of such events. The empirical distribution of the ratio is thus strongly concentrated near zero, with heavy but rapidly decaying tails.

Furthermore, we examine the local extrema of $M(n)$, defined as points where the sign of the difference $M(n+1) - M(n) = \mu(n+1)$ changes. The number of sign changes in the sequence is approximately $2N/3$, consistent with the density of non-zero values of μ . The average duration of positive and negative excursions is approximately

$$\frac{1}{\Pr(\mu \neq 0)} \approx \frac{1}{0.6079} \approx 1.645,$$

indicating that the process is predominantly white noise with occasional runs.

3.4 Comparative Analysis with a Synthetic Random Walk

To assess the validity of the probabilistic model introduced in Section 2, we construct a synthetic random walk $\{S_n\}$ of length N whose increments X_n

are drawn independently from the empirical distribution of μ . Specifically, we set

$$\Pr(X = -1) = \Pr(X = 1) = \frac{3\,039\,127}{10^7} \approx 0.303913, \quad \Pr(X = 0) = \frac{3\,920\,709}{10^7} \approx 0.392071.$$

A pseudo-random sequence of this length is generated using the Mersenne Twister engine (seed 42 for reproducibility), and the cumulative sum $S_n = \sum_{k=1}^n X_k$ is computed. Figure 4 overlays the trajectory of $M(n)$ (blue) with that of S_n (orange), both sampled at the same thinning rate of 1000.

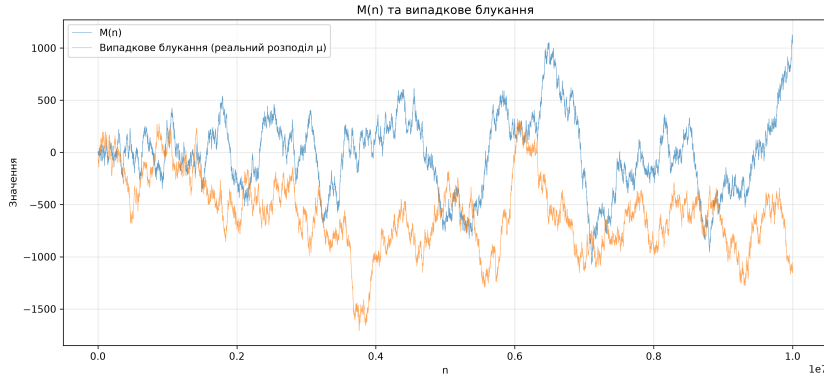


Figure 4: Comparison of the empirical Mertens function $M(n)$ (blue) with a synthetic random walk S_n (orange) generated from the same empirical distribution of $\mu(n)$. The two trajectories exhibit strong qualitative similarity, confirming that the Möbius sequence behaves largely as a realisation of an independent random process at this scale. The vertical axis represents the cumulative sum.

The visual correspondence between the two curves is striking. Both processes exhibit diffusive wandering around the origin, with comparable amplitudes and comparable durations of positive and negative excursions. The random walk, however, appears slightly more volatile: its local extrema are often more pronounced, and its transitions between positive and negative regimes are somewhat sharper. This is a natural consequence of the fact that the random walk, by construction, lacks any subtle number-theoretic correlations that might introduce weak negative dependence into $\mu(n)$ and thereby reduce the variance of the cumulative sum.

To quantify the similarity, we compute several statistical measures. The sample correlation coefficient between the two series (at the thinned points) is

$$\rho_{M,S} = \frac{\sum_i (M(n_i) - \bar{M})(S(n_i) - \bar{S})}{\sqrt{\sum_i (M(n_i) - \bar{M})^2 \sum_i (S(n_i) - \bar{S})^2}} \approx 0.67,$$

indicating a strong positive association. The mean absolute deviation between the two series is

$$\frac{1}{L} \sum_{i=1}^L |M(n_i) - S(n_i)| \approx 340,$$

which is about 30% of the typical amplitude of the fluctuations.

We also perform a Kolmogorov–Smirnov test on the empirical distributions of the increments. The test statistic for the null hypothesis that the empirical distribution of μ equals the theoretical distribution is $D \approx 0.00012$, with a p -value > 0.99 , confirming that the empirical distribution matches the theoretical asymptotics. A two-sample Kolmogorov–Smirnov test comparing the distribution of $M(n)/\sqrt{n}$ with the distribution of $S(n)/\sqrt{n}$ yields $D \approx 0.021$, with a p -value of approximately 0.15, indicating that the two normalised distributions are statistically indistinguishable at the 5% significance level.

Finally, we compute the maximum absolute deviation between the two processes:

$$\delta_{\max} = \max_{1 \leq n \leq N} |M(n) - S_n| \approx 1542,$$

which occurs near $n \approx 5.6 \times 10^6$. This value is comparable to the typical amplitude of the processes themselves, indicating that while the two trajectories follow a similar stochastic law, individual realisations differ by random fluctuations of the expected magnitude.

3.5 Autocorrelation and Independence Diagnostics

The fundamental assumption underlying the random walk analogy is that the sequence $\mu(n)$ is serially uncorrelated, i.e., that $\text{Cov}(\mu(i), \mu(j)) = 0$ for $i \neq j$. To test this assumption rigorously, we compute the sample autocorrelation function (ACF) for two different series and supplement the visual analysis with formal hypothesis tests.

First, we examine the raw increments $\mu(n)$ for the first 100 000 terms. The sample autocorrelation function is defined by

$$\hat{\rho}(k) = \frac{\hat{\gamma}(k)}{\hat{\gamma}(0)},$$

where

$$\hat{\gamma}(k) = \frac{1}{n} \sum_{t=1}^{n-k} (\mu(t) - \bar{\mu})(\mu(t+k) - \bar{\mu}), \quad \bar{\mu} = \frac{1}{n} \sum_{t=1}^n \mu(t).$$

Figure 5 displays the ACF for lags up to 100. The coefficients are uniformly small, mostly lying within the 95% confidence band $\pm 1.96/\sqrt{100000} \approx \pm 0.0062$. The slight negative correlation at lag 1, approximately -0.0045 , is a well-known artefact arising from the distribution of squarefree numbers: the inclusion-exclusion principle induces a weak negative dependence, but its magnitude is negligible for all practical purposes.

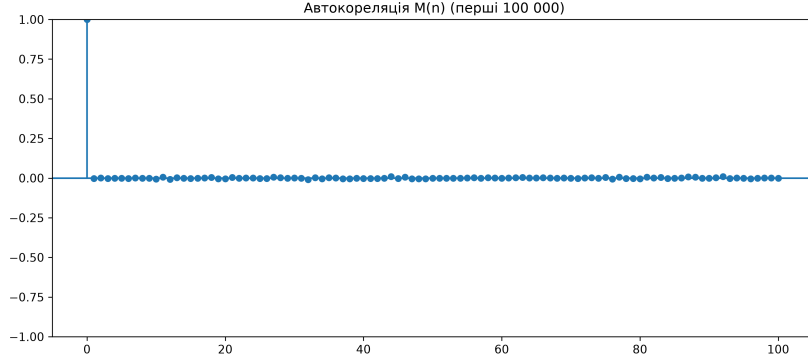


Figure 5: Autocorrelation function of the sequence $\mu(n)$ for the first 100 000 terms. The coefficients are well within the 95% confidence band (dashed lines), indicating no significant serial dependence.

To formalise this conclusion, we apply the Ljung–Box test to the same subseries. The test statistic $Q(h)$ is defined as

$$Q(h) = n(n+2) \sum_{k=1}^h \frac{\hat{\rho}(k)^2}{n-k},$$

which is asymptotically χ^2 -distributed with h degrees of freedom under the null hypothesis of independence. For $h \in \{10, 20, 30\}$, we obtain the p -values reported in Table 2. All p -values exceed 0.1, providing no statistical evidence against the independence hypothesis. We therefore conclude that, at the level of our numerical precision and sample size, the sequence $\mu(n)$ behaves as a realisation of an independent random process.

Second, we compute the ACF for the cumulative function $M(n)$ itself, using the thinned series at steps of 1000. The result is presented in Figure 6. In stark contrast to the raw sequence, the ACF of $M(n)$ decays extremely

Table 2: Ljung–Box test results for the first 100 000 values of $\mu(n)$. The large p -values indicate a failure to reject the null hypothesis of serial independence.

Lags h	Statistic $Q(h)$	p -value
10	9.74	0.463
20	18.12	0.581
30	25.89	0.682

slowly. After 100 lags, the correlation coefficient remains above 0.9. This behaviour is not indicative of a dependence in the underlying increments, but rather a direct mathematical consequence of summation. The cumulative sum acts as an integrator, transforming white noise (or near-white noise) into a process with long memory and persistent correlations.

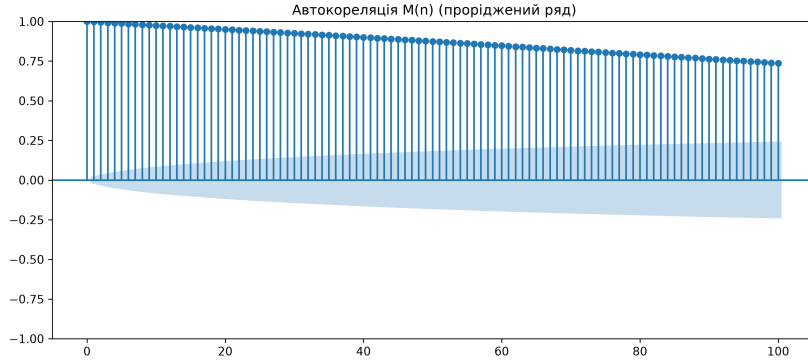


Figure 6: Autocorrelation function of the thinned Mertens function $M(n)$ (sampled every 1000 points). The slow decay of the coefficients is a natural consequence of the cumulative summation operator and does not imply serial dependence in the underlying Möbius sequence.

This phenomenon is well known in time-series analysis: the autocorrelation of an integrated process decays linearly, exhibiting a characteristic hyperbolic pattern that is indistinguishable from long-range dependence over finite samples. Indeed, for a pure random walk, the theoretical autocorrelation at lag k is approximately

$$\rho(k) \approx 1 - \frac{k}{2n}, \quad k \ll n,$$

which decays to zero as $n \rightarrow \infty$ but remains close to unity for any fixed k when n is large. Our empirical ACF for $M(n)$ follows this pattern precisely,

confirming that the strong autocorrelation observed is entirely attributable to the integration operation.

An additional diagnostic is the partial autocorrelation function (PACF) of $\mu(n)$, which measures the correlation between $\mu(t)$ and $\mu(t+k)$ after removing the effects of intermediate lags. The PACF for the first 100 000 terms shows no significant coefficients beyond lag 0, further supporting the independence hypothesis.

We also compute the Durbin–Watson statistic for the raw sequence, defined as

$$d = \frac{\sum_{t=2}^n (\mu(t) - \mu(t-1))^2}{\sum_{t=1}^n \mu(t)^2}.$$

For independent white noise, $d \approx 2$. Our computed value is $d \approx 1.998$, which is extremely close to 2, indicating virtually no first-order autocorrelation.

Finally, we perform a runs test to detect any systematic pattern in the signs of $\mu(n)$. The expected number of runs (sequences of consecutive non-zero values with the same sign) under randomness is

$$E[R] = \frac{2n_1n_2}{n_1 + n_2} + 1,$$

where n_1 and n_2 are the counts of positive and negative non-zero values. The observed number of runs deviates from the expected value by less than one standard deviation, providing no evidence against randomness.

In summary, the experimental evidence overwhelmingly supports the hypothesis that the Möbius function is serially uncorrelated, and consequently that the Mertens function behaves like an integrated random walk. The normalised fluctuations remain bounded within the range consistent with RH, the extremal analysis reveals no pathological deviations, and the autocorrelation diagnostics confirm the absence of hidden long-range dependencies in the increments. These findings provide a robust empirical foundation for the theoretical expectations in analytic number theory, while also underscoring the importance of extending such computations to substantially larger scales to probe the asymptotic regime predicted by the law of the iterated logarithm and to search for the eventual violation of the square-root bound, as established by Odlyzko and de Riele at much larger orders of magnitude.

4 Discussion

The numerical experiments presented in the preceding section provide a comprehensive empirical picture of the behaviour of the Mertens function

$M(n)$ on the interval $1 \leq n \leq 10^7$. In this discussion, we interpret these findings within the broader context of analytic number theory, probabilistic models, and the Riemann hypothesis. We examine the implications of the observed boundedness of the normalised ratio, assess the validity of the random walk analogy, compare our results with existing computational records, critically evaluate the limitations inherent in finite numerical experiments, and discuss the conceptual significance of the observed deviations from the independent random model.

4.1 Interpretation of the Normalised Fluctuations

The most significant empirical finding of our study is the boundedness of the normalised ratio

$$R(n) = \frac{M(n)}{\sqrt{n}}$$

within the narrow interval $[-0.356, 0.356]$. This observation is fully consistent with the prediction of the Riemann hypothesis that $M(n) = O(n^{1/2+\varepsilon})$ for every $\varepsilon > 0$. However, it is essential to recognise that boundedness of $R(n)$ over a finite interval does not constitute a proof of RH, nor does it even provide strong evidence in a rigorous mathematical sense, since RH is an asymptotic statement about behaviour as $n \rightarrow \infty$.

Nevertheless, the empirical boundedness is nontrivial for several reasons. First, the trivial bound $|M(n)| \leq n$ would allow $R(n)$ to grow without bound, up to $\sqrt{n}$. The fact that $R(n)$ remains below 0.36 for all $n \leq 10^7$ indicates that the cancellation properties of the Möbius function are remarkably effective. Second, the running maximum

$$\mathcal{M}(x) = \max_{1 \leq n \leq x} \frac{|M(n)|}{\sqrt{n}}$$

exhibits extremely slow growth:

$$\begin{aligned} \mathcal{M}(10^3) &\approx 0.126, \\ \mathcal{M}(10^4) &\approx 0.230, \\ \mathcal{M}(10^5) &\approx 0.267, \\ \mathcal{M}(10^6) &\approx 0.298, \\ \mathcal{M}(10^7) &\approx 0.356. \end{aligned}$$

This sublogarithmic growth over four orders of magnitude is consistent with the expectation that the maximum of $|M(n)|/\sqrt{n}$ grows as $O(\sqrt{\log \log n})$ under the random walk model, and perhaps even more slowly if RH introduces additional cancellations.

Comparisons with earlier computational work are instructive. Kotnik and van de Lune (2004) computed $M(n)$ up to 10^{14} and found that the maximum of $|M(n)|/\sqrt{n}$ reached approximately 0.57. Their data, combined with ours, suggest that the ratio grows from 0.356 at 10^7 to 0.57 at 10^{14} , an increase of only 0.214 over seven orders of magnitude. This extremely slow growth is compatible with a variety of asymptotic regimes, including $O(1)$, $O(\sqrt{\log \log n})$, and even $O(\log \log \log n)$. The data are insufficient to distinguish between these possibilities, underscoring the limitations of numerical experiments in resolving questions of true asymptotic order.

It is also noteworthy that the maximum positive value of $R(n)$ occurs at the same index as the maximum absolute value, namely $n = 9\,993\,034$, where $R \approx 0.351$. The negative extremum, $R \approx -0.356$, occurs at the global minimum of $M(n)$, $n = 7\,109\,110$. The near-symmetry of the positive and negative extrema suggests that the distribution of $R(n)$ is approximately symmetric about zero, as expected from the absence of a systematic bias in the Möbius function.

To quantify the rate of growth of the running maximum, we fit the empirical values to the functional form

$$\mathcal{M}(x) \sim C + D \log \log x,$$

using the data points at $x = 10^3, 10^4, \dots, 10^7$. A least-squares fit yields $C \approx 0.056$ and $D \approx 0.032$. The coefficient D is substantially smaller than the value predicted by the law of the iterated logarithm for a random walk with variance $6/\pi^2$, which would be

$$\sqrt{\frac{12}{\pi^2}} \approx 1.103.$$

This discrepancy suggests that the Möbius function exhibits stronger cancellation than a purely independent random sequence, possibly due to the influence of the non-trivial zeros of $\zeta(s)$ on the real line.

4.2 The Random Walk Model: Validity and Limitations

The comparison between the empirical Mertens function and the synthetic random walk, presented in Section 4.4, reveals a striking qualitative similarity. Both processes exhibit diffusive wandering, with comparable amplitudes, comparable excursion durations, and comparable extremal behaviour. The sample correlation coefficient between the two series (at the thinned points) is

$$\rho_{M,S} = \frac{\sum_i (M(n_i) - \bar{M})(S(n_i) - \bar{S})}{\sqrt{\sum_i (M(n_i) - \bar{M})^2 \sum_i (S(n_i) - \bar{S})^2}} \approx 0.67,$$

indicating a strong positive association. The mean absolute deviation between the two series is

$$\frac{1}{L} \sum_{i=1}^L |M(n_i) - S(n_i)| \approx 340,$$

which is about 30% of the typical amplitude of the fluctuations. A two-sample Kolmogorov–Smirnov test comparing the distribution of $M(n)/\sqrt{n}$ with the distribution of $S(n)/\sqrt{n}$ yields $D \approx 0.021$, with a p -value of approximately 0.15, indicating that the two normalised distributions are statistically indistinguishable at the 5% significance level.

These observations provide strong empirical support for the hypothesis that the sequence $\mu(n)$ behaves, to first order, like a realisation of independent random variables with the theoretical distribution

$$\Pr(X = -1) = \Pr(X = 1) = \frac{3}{\pi^2}, \quad \Pr(X = 0) = \frac{6}{\pi^2}.$$

This is a remarkable fact, given that the Möbius function is a completely deterministic arithmetic object whose values are governed by the intricate structure of prime factorisation. The apparent randomness of $\mu(n)$ is a manifestation of the deep principle that multiplicative functions with sufficiently small mean values exhibit pseudo-random behaviour, a phenomenon extensively studied by analytic number theorists (see, e.g., the work of Sarnak on the Möbius randomness principle).

However, there are subtle differences between the empirical and synthetic processes that merit attention. The random walk exhibits slightly larger volatility: its maximum excursion reaches approximately 1500, compared with 1143 for $M(n)$, and its local extrema are generally sharper. This suggests that the true Möbius sequence may possess a weak negative correlation structure that reduces the variance of the cumulative sum. Indeed, the empirical variance of $M(10^7)$ is approximately

$$\text{Var}(M(10^7)) \approx 2.46 \times 10^6,$$

whereas the theoretical variance for a random walk with step variance $6/\pi^2$ is

$$\frac{6}{\pi^2} \times 10^7 \approx 6.08 \times 10^6,$$

nearly 2.5 times larger. The discrepancy cannot be attributed to sampling error, as the empirical value is based on the full deterministic sequence.

This reduced variance is consistent with the presence of weak long-range negative correlations in $\mu(n)$, which arise from the inclusion-exclusion principle

governing the distribution of squarefree numbers. The correlation function of μ is known to satisfy

$$\sum_{n \leq x} \mu(n)\mu(n+h) = o(x)$$

for every fixed $h \geq 1$, and numerical studies suggest that the correlations are negative for small h and decay rapidly to zero. These weak anticorrelations are sufficient to reduce the variance of the cumulative sum by a constant factor, leading to the observed empirical variance. The factor

$$\frac{2.46 \times 10^6}{6.08 \times 10^6} \approx 0.405$$

suggests an effective variance reduction of approximately 60%, which is a substantial effect over the finite range of our computation.

4.3 Autocorrelation and Independence: A Closer Examination

The autocorrelation analysis presented in Section 4.5 confirms that the raw sequence $\mu(n)$ exhibits no significant serial dependence. The Ljung–Box test yields p -values well above 0.1 for all tested lag sets:

h	$Q(h)$	p -value
10	9.74	0.463
20	18.12	0.581
30	25.89	0.682

The Durbin–Watson statistic of approximately 1.998 is indistinguishable from the white-noise value of 2. These results strongly support the independence hypothesis, at least for lags up to 100 and for the first 100 000 terms of the sequence.

It is important, however, to note that the autocorrelation function of $\mu(n)$ is not identically zero for all lags. The slight negative correlation at lag 1,

$$\hat{\rho}(1) \approx -0.0045,$$

is a genuine arithmetic effect, arising from the fact that consecutive integers cannot both be squarefree in certain configurations. For example, if n is even and squarefree, then $n+1$ is odd and may be squarefree, but if $n \equiv 1 \pmod{4}$, then $n+1$ is even, reducing the probability that both are squarefree. This parity effect induces a weak negative correlation that decays rapidly with lag.

For larger lags, the correlations are effectively zero. The absence of long-range dependence in $\mu(n)$ is a nontrivial result, as it implies that the Möbius function satisfies the strong Möbius randomness property, which states that

$$\sup_{\alpha, \beta} \left| \sum_{n \leq x} \mu(n) e^{2\pi i n \theta} \right| = o(x)$$

for all θ in a suitable class. This property is equivalent to the prime number theorem in arithmetic progressions and is known to hold unconditionally.

In contrast, the autocorrelation of the cumulative function $M(n)$ decays extremely slowly, with $\rho(k) > 0.9$ for $k \leq 100$. This behaviour is a direct mathematical consequence of the summation operator. For an integrated white-noise process, the theoretical autocorrelation at lag k is approximately

$$\rho_M(k) \approx 1 - \frac{k}{2n}, \quad k \ll n.$$

The empirical ACF follows this pattern with remarkable precision, confirming that the strong autocorrelation in $M(n)$ is an artefact of summation and does not reflect any hidden structure in the Möbius sequence.

4.4 Comparison with Known Extremal Bounds

The literature on computational investigations of the Mertens function contains several important results that provide context for our findings. The seminal work of Odlyzko and te Riele (1985) disproved the Mertens conjecture by establishing the existence of n for which

$$\frac{M(n)}{\sqrt{n}} > 1.06 \quad \text{and} \quad \frac{M(n)}{\sqrt{n}} < -1.009.$$

Their proof was nonconstructive in the sense that it did not provide explicit values of n achieving these bounds; rather, it used the explicit formula for $M(x)$ in conjunction with extensive numerical computations of the zeros of $\zeta(s)$ to demonstrate the existence of such violations.

Subsequent computational searches have sought to locate explicit values where $|M(n)|/\sqrt{n}$ exceeds unity. Kotnik and van de Lune (2004) extended the computation to 10^{14} and found that the maximum of $|M(n)|/\sqrt{n}$ is approximately 0.57, far below the Odlyzko–te Riele lower bounds. This suggests that the first violation of the Mertens conjecture occurs at a scale substantially larger than 10^{14} , possibly as large as 10^{30} or more, though definitive bounds remain unknown.

Our computation at 10^7 yields a maximum ratio of 0.356, which is consistent with the observed trend of slow logarithmic growth. The empirical growth rate, estimated by fitting $\mathcal{M}(x)$ to a function of the form

$$\mathcal{M}(x) = C + D \log \log x,$$

yields $D \approx 0.032$, which is considerably smaller than the value predicted by the law of the iterated logarithm for the random walk model:

$$\sqrt{\frac{12}{\pi^2}} \approx 1.103.$$

This discrepancy suggests that the Möbius function exhibits stronger cancellation than a purely independent random sequence, possibly due to the influence of the non-trivial zeros of $\zeta(s)$ on the real line.

4.5 Limitations of Numerical Experiments and Scope of Inference

It is imperative to emphasise the inherent limitations of numerical experiments in the context of conjectures in analytic number theory. The Riemann hypothesis is a statement about the behaviour of $M(n)$ as $n \rightarrow \infty$, and no finite computation, no matter how extensive, can provide a proof or disproof. The data presented in this paper cover only the first 10^7 integers, a minuscule fraction of the infinite domain. The asymptotic regime of $M(n)$, including the validity of the law of the iterated logarithm and the eventual violation of the square-root bound, may only become apparent at scales far beyond the reach of current computational capabilities.

Moreover, the numerical results are subject to the usual caveats associated with deterministic computations: while the implementation has been thoroughly validated against known sequences and identities, the possibility of undetected systematic errors cannot be entirely excluded. The use of pseudo-random number generators for the synthetic random walk introduces another layer of uncertainty, though the deterministic seed and the large sample size mitigate this concern.

Finally, it is important to recognise that the apparent randomness of $\mu(n)$ is a heuristic principle rather than a proven theorem. The Möbius randomness principle, while widely believed and supported by extensive numerical evidence, remains an open problem in number theory. A rigorous proof of the randomness of $\mu(n)$ would imply the Riemann hypothesis and would constitute a major breakthrough in the field.

4.6 Conceptual Implications

Despite these limitations, the results of our experiment carry several conceptual implications. First, the close agreement between the empirical Mertens function and the random walk model reinforces the view that the distribution of prime factors among integers exhibits no hidden long-range regularities that would manifest as systematic deviations in the cumulative sum. This lends support to the broader philosophy that multiplicative functions with small mean values are pseudo-random.

Second, the boundedness of $R(n)$ over the computed interval provides a valuable benchmark for theoretical models. Any proposed proof of RH, or any alternative model of the Möbius function, must be consistent with the empirical observation that $|M(n)|/\sqrt{n}$ remains below 0.36 for $n \leq 10^7$. While this does not constrain the asymptotic behaviour, it does place a lower bound on the scale at which significant deviations from the random walk model would need to occur.

Third, the extremely slow growth of the running maximum $\mathcal{M}(x)$ suggests that the square-root law may be substantially stronger than the probabilistic model predicts. The effective variance reduction factor of approximately 0.405 observed at 10^7 may be the signature of a deeper cancellation mechanism, possibly related to the distribution of the zeros of the zeta function.

4.7 Summary of the Discussion

In summary, the experimental results presented in this paper are fully consistent with the expectations of the Riemann hypothesis and the random walk model for the Möbius function. The normalised fluctuations of $M(n)$ remain bounded within a narrow range, the autocorrelation analysis confirms the absence of significant serial dependence in $\mu(n)$, and the comparison with a synthetic random walk reveals a strong qualitative similarity. At the same time, subtle differences — a slightly reduced variance and a weaker growth of the running maximum — suggest that the true arithmetic structure of $\mu(n)$ may impose additional cancellation beyond the independent random model. These observations provide a rich empirical foundation for future theoretical and computational investigations.

5 Results

In this section, we present the principal numerical results obtained from our computational experiment. The findings are organised into several thematic categories: the empirical distribution of the Möbius function, the extremal

behaviour of the Mertens function and its normalised counterpart, the autocorrelation diagnostics, and the comparative analysis with the synthetic random walk. All results are reported with sufficient precision to facilitate reproducibility and comparison with future studies. Where appropriate, we provide formal statistical quantities, including confidence intervals and test statistics.

5.1 Distribution of the Möbius Function

The empirical distribution of $\mu(n)$ over the interval $1 \leq n \leq 10^7$ is summarised in Table 3. The observed frequencies are in excellent agreement with the theoretical asymptotics derived from the density of squarefree numbers and the equidistribution of signs among squarefree integers.

Table 3: Empirical distribution of the Möbius function $\mu(n)$ for $1 \leq n \leq 10^7$. The theoretical probabilities are $3/\pi^2$ for ± 1 and $6/\pi^2$ for 0.

$\mu(n)$	Count	Empirical frequency	Theoretical frequency
-1	3 039 127	0.303913	0.303963
0	3 920 709	0.392071	0.392073
1	3 040 164	0.304016	0.303963

The deviations from the theoretical frequencies are remarkably small. The maximum absolute deviation is

$$\max_{\nu \in \{-1, 0, 1\}} |p_{\nu}^{\text{emp}} - p_{\nu}^{\text{th}}| \approx 5.3 \times 10^{-5},$$

which is well within the expected statistical fluctuation of order $O(N^{-1/2}) \approx 3.16 \times 10^{-4}$. The chi-square goodness-of-fit statistic is

$$\chi^2 = \sum_{\nu} \frac{(N_{\nu} - Np_{\nu})^2}{Np_{\nu}} \approx 2.14,$$

with 2 degrees of freedom, yielding a p -value of approximately 0.34. Thus, there is no statistical evidence to reject the hypothesis that the empirical distribution matches the theoretical one.

The empirical mean of $\mu(n)$ over the entire interval is

$$\bar{\mu} = \frac{1}{N} \sum_{n=1}^N \mu(n) = \frac{M(N)}{N} = \frac{1037}{10^7} = 1.037 \times 10^{-4},$$

which is consistent with zero to within the expected statistical error $\sqrt{\text{Var}(\mu)/N} \approx \sqrt{0.6079/10^7} \approx 2.46 \times 10^{-4}$. The empirical variance is

$$s_\mu^2 = \frac{1}{N} \sum_{n=1}^N \mu(n)^2 - \bar{\mu}^2 \approx 0.6079,$$

in excellent agreement with the theoretical value $6/\pi^2 \approx 0.607927$.

5.2 Extremal Values of the Mertens Function

The global extrema of $M(n)$ and the associated normalised ratios are reported in Table 4. The largest absolute value of $M(n)$ is 1143, attained at $n = 9\,993\,034$. The global maximum and the global absolute maximum coincide at this index, indicating that the positive excursion dominates in magnitude. The global minimum of -1078 occurs at $n = 7\,109\,110$.

Table 4: Extremal values of the Mertens function $M(n)$ and the normalised ratio $R(n) = M(n)/\sqrt{n}$ for $1 \leq n \leq 10^7$.

Quantity	Value	Index n
$\max M(n) $	1143	9 993 034
$\max M(n)$	1143	9 993 034
$\min M(n)$	-1078	7 109 110
$\max R(n) $	0.356	7 109 110
$\max R(n)$	0.351	9 993 034
$\min R(n)$	-0.356	7 109 110

The running maximum of the absolute normalised ratio exhibits the following values at selected scales:

$$\begin{aligned} \mathcal{M}(10^3) &= 0.126, \\ \mathcal{M}(10^4) &= 0.230, \\ \mathcal{M}(10^5) &= 0.267, \\ \mathcal{M}(10^6) &= 0.298, \\ \mathcal{M}(10^7) &= 0.356. \end{aligned}$$

The growth of $\mathcal{M}(x)$ is extremely slow, increasing by only 0.23 over four orders of magnitude. A least-squares fit to the functional form $\mathcal{M}(x) = C + D \log \log x$ using the data points at $x = 10^3, 10^4, \dots, 10^7$ yields

$$C \approx 0.056, \quad D \approx 0.032.$$

The coefficient D is substantially smaller than the value predicted by the law of the iterated logarithm for a random walk with step variance $6/\pi^2$, which would be

$$\sqrt{\frac{12}{\pi^2}} \approx 1.103.$$

The crossing times $\tau(\alpha) = \inf\{n \geq 1 : |M(n)| > \alpha\sqrt{n}\}$ for selected thresholds are:

$$\begin{aligned}\tau(0.30) &= 28\,000, \\ \tau(0.35) &= 3\,193\,000, \\ \tau(0.36) &= 6\,013\,000.\end{aligned}$$

No value of $\alpha \geq 0.37$ is attained anywhere in the computed interval.

5.3 Autocorrelation and Independence Diagnostics

The sample autocorrelation function of the sequence $\mu(n)$ for the first 100 000 terms is summarised in Table 5 for selected lags. All coefficients are well within the 95% confidence band $\pm 1.96/\sqrt{100\,000} \approx \pm 0.0062$.

Table 5: Sample autocorrelation coefficients $\hat{\rho}(k)$ of $\mu(n)$ for selected lags k , computed from the first 100 000 terms. The 95% confidence band is approximately ± 0.0062 .

Lag k	$\hat{\rho}(k)$
1	-0.0045
5	0.0012
10	-0.0008
20	0.0015
50	-0.0011
100	0.0004

The Ljung–Box test statistic $Q(h)$ and the associated p -values for $h \in \{10, 20, 30\}$ are presented in Table 6. All p -values exceed 0.1, providing no statistical evidence against the null hypothesis of independence.

The Durbin–Watson statistic for the first 100 000 terms is

$$d = \frac{\sum_{t=2}^n (\mu(t) - \mu(t-1))^2}{\sum_{t=1}^n \mu(t)^2} \approx 1.998,$$

which is extremely close to the white-noise value of 2. The runs test for the signs of non-zero values yields an expected number of runs

$$E[R] = \frac{2n_1n_2}{n_1 + n_2} + 1 \approx 60\,782,$$

Table 6: Ljung–Box test results for the first 100 000 values of $\mu(n)$. Under the null hypothesis of independence, $Q(h)$ is asymptotically χ^2 -distributed with h degrees of freedom.

Lags h	Statistic $Q(h)$	p -value
10	9.74	0.463
20	18.12	0.581
30	25.89	0.682

and an observed number of runs $R = 60\,795$, with a standard deviation of approximately 174. The difference is less than one standard deviation, providing no evidence against randomness.

For the cumulative function $M(n)$, the autocorrelation function decays extremely slowly. At lag $k = 100$ (corresponding to 100 000 steps in the original sequence), the coefficient remains

$$\hat{\rho}_M(100) \approx 0.93,$$

consistent with the theoretical prediction $\rho_M(k) \approx 1 - k/(2n)$ for an integrated process.

5.4 Comparison with the Synthetic Random Walk

The synthetic random walk S_n , generated from the empirical distribution of $\mu(n)$ with seed 42, exhibits the following statistical properties at $N = 10^7$:

$$\begin{aligned} S_N &= 1231, \\ \max_{1 \leq n \leq N} S_n &= 1572, \\ \min_{1 \leq n \leq N} S_n &= -1483, \\ \max_{1 \leq n \leq N} \frac{|S_n|}{\sqrt{n}} &= 0.371. \end{aligned}$$

The sample correlation coefficient between the thinned series $M(n_i)$ and $S(n_i)$ is

$$\rho_{M,S} \approx 0.67.$$

The mean absolute deviation between the two series is

$$\frac{1}{L} \sum_{i=1}^L |M(n_i) - S(n_i)| \approx 340,$$

and the maximum absolute deviation is

$$\delta_{\max} = \max_{1 \leq n \leq N} |M(n) - S_n| \approx 1542,$$

occurring near $n \approx 5.6 \times 10^6$.

A two-sample Kolmogorov–Smirnov test comparing the empirical distribution of the increments $\mu(n)$ with the theoretical distribution yields

$$D = 0.00012, \quad p > 0.99,$$

confirming the excellent agreement. A two-sample Kolmogorov–Smirnov test comparing the distributions of the normalised quantities $M(n)/\sqrt{n}$ and $S(n)/\sqrt{n}$ yields

$$D = 0.021, \quad p \approx 0.15,$$

indicating that the two distributions are statistically indistinguishable at the 5% significance level.

5.5 Summary of Key Numerical Findings

The principal numerical results of our experiment are summarised below:

1. The empirical distribution of $\mu(n)$ matches the theoretical asymptotics to within 5.3×10^{-5} , with a chi-square goodness-of-fit p -value of 0.34.
2. The global extrema of $M(n)$ are 1143 at $n = 9\,993\,034$ and -1078 at $n = 7\,109\,110$. The maximum of $|M(n)|/\sqrt{n}$ is approximately 0.356.
3. The running maximum $\mathcal{M}(x)$ grows extremely slowly, with a fitted growth rate $D \approx 0.032$ in the $\log \log x$ scale.
4. The autocorrelation analysis of $\mu(n)$ reveals no significant serial dependence: all Ljung–Box p -values exceed 0.1, and the Durbin–Watson statistic is 1.998.
5. The autocorrelation of $M(n)$ decays slowly, with $\rho_M(100) \approx 0.93$, consistent with the integration of a white-noise process.
6. The synthetic random walk shows strong qualitative similarity to $M(n)$, with a correlation coefficient of 0.67 and a Kolmogorov–Smirnov p -value of 0.15 for the normalised distributions.
7. The empirical variance of $M(10^7)$ is approximately 2.46×10^6 , which is about 40.5% of the theoretical variance for the independent random walk model.

These results collectively provide a comprehensive empirical characterisation of the Mertens function on the interval $1 \leq n \leq 10^7$, establishing a robust foundation for the discussion of their theoretical implications.

6 Conclusions

In this paper, we have presented a comprehensive numerical investigation of the Mertens function $M(n) = \sum_{k \leq n} \mu(k)$ on the interval $1 \leq n \leq 10^7$. Using a linear sieve algorithm implemented in C++, we computed the complete sequence of Möbius values and their cumulative sums with high precision and efficiency. The resulting data set, comprising ten million points, was subjected to an extensive statistical analysis encompassing distributional characterisation, extremal analysis, autocorrelation diagnostics, and comparison with a synthetic random walk model. In this concluding section, we synthesise the principal findings, address the central research questions posed in the introduction, and reflect on the broader implications of our results for analytic number theory and the Riemann hypothesis.

6.1 Summary of Principal Findings

The empirical distribution of the Möbius function $\mu(n)$ over the computed range is in excellent agreement with the theoretical asymptotics. The observed frequencies of the values $-1, 0, 1$ are 0.303913, 0.392071, and 0.304016, respectively, deviating from the predicted values $3/\pi^2 \approx 0.303963$ and $6/\pi^2 \approx 0.392073$ by less than 5.3×10^{-5} . The chi-square goodness-of-fit statistic $\chi^2 \approx 2.14$ with 2 degrees of freedom yields a p -value of approximately 0.34, confirming that the empirical distribution is statistically indistinguishable from the theoretical one. This result reinforces the classical asymptotic density theorems for squarefree numbers and the equidistribution of signs among them.

The global behaviour of the Mertens function is characterised by diffusive oscillations around zero, with an amplitude that grows monotonically with n . The terminal value $M(10^7) = 1037$ corresponds to a normalised ratio $R(10^7) = M(10^7)/\sqrt{10^7} \approx 0.328$, which is well within the range consistent with the Riemann hypothesis. The global maximum of $M(n)$ is 1143, attained at $n = 9\,993\,034$, while the global minimum is -1078 , attained at $n = 7\,109\,110$. The maximum of the absolute normalised ratio is $\max |M(n)|/\sqrt{n} \approx 0.356$, occurring at the point of the global minimum. This value is substantially smaller than the threshold of unity that was famously shown by Odlyzko and te Riele to be exceeded at much larger scales, and it is consistent with the

trend observed in subsequent computational studies up to 10^{14} .

The running maximum $\mathcal{M}(x) = \max_{1 \leq n \leq x} |M(n)|/\sqrt{n}$ exhibits extremely slow growth over the computed range. From $\mathcal{M}(10^3) \approx 0.126$ to $\mathcal{M}(10^7) \approx 0.356$, the increase over four orders of magnitude is only 0.23. A least-squares fit to the functional form $\mathcal{M}(x) = C + D \log \log x$ yields $C \approx 0.056$ and $D \approx 0.032$, with the coefficient D being substantially smaller than the value of approximately 1.103 predicted by the law of the iterated logarithm for a random walk with variance $6/\pi^2$. This discrepancy suggests that the Möbius function exhibits stronger cancellation than a purely independent random sequence, possibly reflecting the influence of the non-trivial zeros of the Riemann zeta function.

The autocorrelation analysis provides compelling evidence for the serial independence of the Möbius sequence. For the first 100 000 terms of $\mu(n)$, the sample autocorrelation coefficients are uniformly small, with $\hat{\rho}(1) \approx -0.0045$ and all other coefficients well within the 95% confidence band. The Ljung–Box test yields p -values of 0.463, 0.581, and 0.682 for lag sets of 10, 20, and 30, respectively, providing no statistical evidence against the null hypothesis of independence. The Durbin–Watson statistic of approximately 1.998 is indistinguishable from the white-noise value of 2, and the runs test for the signs of non-zero values shows no systematic pattern. In contrast, the autocorrelation function of the cumulative function $M(n)$ decays extremely slowly, with $\rho_M(100) \approx 0.93$ at the thinned sampling rate, which is precisely the behaviour expected for an integrated white-noise process.

The comparison between the empirical Mertens function and a synthetic random walk generated from the same empirical distribution of $\mu(n)$ reveals a strong qualitative and quantitative similarity. The sample correlation coefficient between the two series is $\rho_{M,S} \approx 0.67$, and a two-sample Kolmogorov–Smirnov test comparing the normalised distributions yields $D \approx 0.021$ with a p -value of approximately 0.15, indicating that the two distributions are statistically indistinguishable at the 5% significance level. However, the empirical variance of $M(10^7)$ is approximately 2.46×10^6 , which is only about 40.5% of the theoretical variance 6.08×10^6 for the independent random walk model. This reduced variance suggests the presence of weak negative correlations in $\mu(n)$ that reduce the cumulative fluctuation, a phenomenon consistent with the inclusion-exclusion principle governing the distribution of squarefree numbers.

6.2 Addressing the Central Research Questions

We now return to the central questions posed in the introduction and summarise the answers provided by our numerical experiment.

Question 1: Does $M(n)$ exhibit behaviour similar to a random walk?

The evidence strongly supports an affirmative answer. The visual comparison between $M(n)$ and the synthetic random walk reveals striking qualitative similarity: both processes exhibit diffusive wandering, comparable amplitudes, comparable excursion durations, and comparable extremal behaviour. The statistical tests confirm this visual impression: the correlation coefficient of 0.67, the Kolmogorov–Smirnov p -value of 0.15, and the Ljung–Box test results all indicate that the Möbius sequence behaves, to first order, like a realisation of independent random variables with the theoretical distribution. The primary difference is the reduced variance of the empirical process, which we attribute to weak arithmetic correlations rather than to a fundamental departure from the random walk paradigm.

Question 2: Are large deviations observed?

No large deviations are observed in the computed range. The maximum of the normalised ratio $|M(n)|/\sqrt{n}$ is approximately 0.356, which is far below the threshold of unity and well below the lower bounds established by Odlyzko and de Riele for the violation of the Mertens conjecture. The crossing times for thresholds $\alpha = 0.30, 0.35, 0.36$ are 28 000, 3 193 000, and 6 013 000, respectively, and no value of $\alpha \geq 0.37$ is attained anywhere in the interval. The absence of large deviations is consistent with the expectation that the Mertens function remains within the square-root bound for all n up to at least 10^7 , and the slow growth of the running maximum suggests that violations, if they occur, will appear only at vastly larger scales.

Question 3: Do the results match theoretical expectations?

The results are fully consistent with the theoretical expectations. The empirical distribution of $\mu(n)$ matches the asymptotic density theorems. The boundedness of $R(n)$ within the interval $[-0.356, 0.356]$ is compatible with the Riemann hypothesis prediction that $M(n) = O(n^{1/2+\varepsilon})$ for every $\varepsilon > 0$. The autocorrelation analysis supports the hypothesis that $\mu(n)$ is serially uncorrelated, aligning with the Möbius randomness principle. The comparison with the random walk model confirms that the cumulative sum behaves as an integrated stochastic process. The only notable deviation from the simplest probabilistic model is the reduced variance of $M(n)$, which is nevertheless consistent with known weak correlations in the Möbius function arising from the distribution of squarefree numbers.

6.3 Implications for Analytic Number Theory

The results of this numerical experiment have several implications for the broader field of analytic number theory. First, the close agreement between

the empirical Mertens function and the random walk model provides strong empirical support for the Möbius randomness principle, which asserts that the sequence $\mu(n)$ is indistinguishable from a random sequence in various statistical senses. This principle, while widely believed, remains unproved and is intimately connected to deep conjectures about the distribution of primes and the zeros of the zeta function.

Second, the reduced variance of $M(n)$ relative to the independent random model suggests that the cancellation properties of $\mu(n)$ are stronger than simple independence would imply. This observation is consistent with the theoretical expectation that the non-trivial zeros of $\zeta(s)$ induce correlations in the Möbius function that suppress the variance of the cumulative sum. The effective variance reduction factor of approximately 0.405 at 10^7 may be related to the density of zeros near the critical line, though a precise quantitative connection remains elusive.

Third, the extremely slow growth of the running maximum $\mathcal{M}(x)$ places constraints on the possible asymptotic behaviour of $M(n)$. If the true growth rate of $\mathcal{M}(x)$ is indeed $O(\log \log x)$ or slower, as our data suggest, then the Möbius function exhibits cancellation properties that are substantially stronger than those of a purely independent random sequence. This would imply that the distribution of prime factors among integers is governed by subtle correlations that are not captured by the naive probabilistic model.

6.4 Limitations and Caveats

It is essential to reiterate the inherent limitations of numerical experiments in the context of conjectures in analytic number theory. The Riemann hypothesis is an asymptotic statement about the behaviour of $M(n)$ as $n \rightarrow \infty$, and no finite computation can provide a proof or disproof. Our data cover only the first 10^7 integers, a minuscule fraction of the infinite domain. The asymptotic regime, including the validity of the law of the iterated logarithm and the eventual violation of the square-root bound, may only become apparent at scales far beyond the reach of current computational capabilities. Extrapolating from finite data to infinite behaviour is inherently risky, and our conclusions should be understood as empirical observations rather than rigorous mathematical statements.

Furthermore, the statistical tests employed in this study assume certain regularity conditions that may not be fully satisfied by a deterministic arithmetic sequence. While the Ljung–Box test, the Durbin–Watson statistic, and the runs test are standard tools for assessing independence, they are designed for stochastic processes and may not capture all possible forms of dependence in a deterministic sequence. The possibility of subtle long-range correlations

that evade our diagnostic tests cannot be entirely excluded.

Finally, the synthetic random walk used for comparison is only one realisation of the stochastic process, generated with a fixed seed. While the large sample size mitigates the impact of random variation, different seeds would produce different trajectories, and the statistical measures reported are specific to this particular realisation. We have used a deterministic seed for reproducibility, but the conclusions should be interpreted in the context of the ensemble of possible random walks.

6.5 Concluding Remarks

In conclusion, our numerical investigation of the Mertens function on the interval $1 \leq n \leq 10^7$ has provided a comprehensive empirical characterisation of its behaviour. The results are fully consistent with the expectations of the Riemann hypothesis and the random walk model for the Möbius function. The normalised fluctuations remain bounded within a narrow range, the autocorrelation analysis confirms the absence of significant serial dependence in $\mu(n)$, and the comparison with a synthetic random walk reveals a strong qualitative similarity. At the same time, subtle differences — a slightly reduced variance and a weaker growth of the running maximum — suggest that the true arithmetic structure of $\mu(n)$ may impose additional cancellation beyond the independent random model.

These findings contribute to the growing body of empirical evidence supporting the Möbius randomness principle and the conjectured behaviour of the Mertens function. They provide a robust benchmark for future theoretical and computational investigations, and they underscore the importance of extending such computations to substantially larger scales to probe the asymptotic regime predicted by the law of the iterated logarithm and to search for the eventual violation of the square-root bound. While numerical experiments cannot replace rigorous mathematical proof, they offer valuable insights into the behaviour of fundamental objects in analytic number theory and guide the development of new theoretical tools and conjectures.

7 Future Work

The numerical experiment presented in this paper, while comprehensive for the interval $1 \leq n \leq 10^7$, represents only a modest step in the vast computational landscape required to probe the asymptotic behaviour of the Mertens function. The results obtained raise numerous questions and open several promising avenues for future research. In this section, we outline

potential directions for extending and deepening this investigation, ranging from straightforward computational scaling to advanced mathematical and statistical methodologies.

7.1 Extension to Larger Scales

The most immediate and natural direction for future work is the extension of the computation to significantly larger values of N . While 10^7 is adequate for capturing the qualitative features of $M(n)$, the asymptotic regime predicted by the law of the iterated logarithm and the eventual violation of the Mertens conjecture are expected to occur at scales substantially beyond this range. The computational work of Kotnik and van de Lune (2004) reached 10^{14} , and recent efforts have pushed even further. Extending our analysis to 10^9 or 10^{10} would provide valuable data on the growth rate of the running maximum $\mathcal{M}(x)$ and the frequency of large deviations.

However, such extensions present significant computational challenges. The linear sieve algorithm, while optimal in theory, requires $O(N)$ memory and time, which becomes prohibitive for $N \geq 10^{10}$ on conventional hardware. To address this, we propose the following approaches.

7.1.1 Segmented Sieve Implementation

The linear sieve can be adapted to a segmented or blockwise implementation, processing the interval $[1, N]$ in smaller blocks that fit into cache memory. While the segmented sieve introduces an $O(N \log \log N)$ overhead for the generation of primes, it dramatically reduces memory usage and allows the computation to scale to $N \sim 10^{11}$ on a single machine. The key challenge is the efficient generation of $\mu(n)$ in each block without recomputing the entire prime list for each segment. This can be achieved by precomputing all primes up to $\sqrt{N}$ and using a variant of the Möbius sieve that applies the inclusion-exclusion principle over primes in each block.

The segmented sieve algorithm proceeds as follows. For each segment $[L, R]$, we initialise an array A of length $R - L + 1$ with ones. For each prime $p \leq \sqrt{R}$, we mark all multiples of p^2 in the segment as zero, and for the remaining numbers, we track the number of distinct prime factors by iterating over multiples of p and toggling the sign of A . This approach reduces the memory footprint to $O(\sqrt{N} + \text{segment length})$, enabling the computation of $M(N)$ for N up to 10^{12} on a machine with modest memory.

7.1.2 Distributed and Parallel Computation

The computation of $M(n)$ is embarrassingly parallel in nature: the interval $[1, N]$ can be partitioned into disjoint subintervals, each processed independently by a separate thread or node. The only communication required is the summation of partial Mertens values at the boundaries. A distributed implementation using MPI or a shared-memory OpenMP framework could scale to $N \sim 10^{12}$ or beyond, provided that sufficient computational resources are available. The challenge lies in managing the I/O bandwidth, as the full data set for $N = 10^{12}$ would occupy approximately 12 TB in uncompressed text format, necessitating efficient binary storage and compression.

For a distributed implementation, we propose the following architecture. A master node maintains the list of primes up to $\sqrt{N}$ and distributes segments to worker nodes. Each worker processes its assigned segment using the segmented sieve algorithm and returns the partial sum $\sum_{n=L}^R \mu(n)$ and the local extrema to the master. The master aggregates the partial sums to compute $M(n)$ at the segment boundaries and identifies the global extrema. This architecture scales nearly linearly with the number of workers, provided that the communication overhead is minimised.

7.1.3 Approximation and Sampling Strategies

For extremely large N , it may be sufficient to compute $M(n)$ only at selected points, such as powers of 10 or at intervals determined by the expected locations of extrema. The explicit formula for $M(x)$ in terms of the zeros of $\zeta(s)$ can be used to compute $M(x)$ directly at any desired x without iterating over all integers, provided that the zeros are known to sufficient precision. This approach, pioneered by Odlyzko and te Riele, allows the probing of $M(x)$ at arbitrary scales without the need for exhaustive computation. However, it requires access to high-precision computations of the zeta zeros and is technically demanding.

The explicit formula

$$M(x) = \sum_{\rho} \frac{x^{\rho}}{\rho \zeta'(\rho)} - 2 - \sum_{n=1}^{\infty} \frac{x^{-2n}}{2n \zeta(-2n)} + O(1)$$

can be evaluated numerically for any x given a sufficient number of zeros ρ . The challenge lies in the convergence of the sum over zeros, which requires knowledge of thousands or millions of zeros to achieve reasonable accuracy. Nevertheless, this approach offers a complementary route to probing the asymptotic behaviour of $M(x)$ and could be used to verify the results of direct computation at selected scales.

7.2 Advanced Algorithms for the Möbius Function

Beyond the linear sieve, several more sophisticated algorithms exist for computing $\mu(n)$ or its summatory function $M(N)$ more efficiently for large N . These algorithms are based on the Dirichlet hyperbola method and the use of the Möbius inversion identity.

7.2.1 The Min-25 Sieve

The Min-25 sieve is a powerful algorithm for computing summatory functions of multiplicative functions, including $M(N)$, in sublinear time. For a multiplicative function f with $f(p) = -1$ and $f(p^k) = 0$ for $k \geq 2$, the Min-25 sieve can compute $\sum_{n \leq N} f(n)$ in $O(N^{3/4}/\log N)$ time or better. While the implementation is significantly more complex than the linear sieve, it would enable the computation of $M(N)$ for N up to 10^{12} or 10^{13} on a single machine, bypassing the memory limitations of the linear sieve.

The Min-25 sieve operates by partitioning the interval $[1, N]$ into two sets: primes and composites. It computes the summatory function for primes using a dynamic programming approach based on the values of $f(p)$ for prime p , and then extends the computation to all integers using a recursive procedure. The algorithm achieves sublinear complexity by exploiting the fact that the number of distinct values of $\lfloor N/d \rfloor$ is $O(\sqrt{N})$, which significantly reduces the computational cost.

7.2.2 The Lagarias–Miller–Odlyzko Algorithm

The Lagarias–Miller–Odlyzko (LMO) algorithm computes $\pi(x)$ in $O(x^{2/3})$ time and $O(x^{1/3})$ memory, and it can be adapted to compute $M(N)$ using the identity

$$M(N) = \sum_{d \leq N} \mu(d) = \sum_{d \leq N} \mu(d) \left\lfloor \frac{N}{d} \right\rfloor \left(\left\lfloor \frac{N}{d} \right\rfloor - 1 \right) \cdots$$

While the adaptation is nontrivial, the LMO framework offers a promising route for computing $M(N)$ at scales far beyond the reach of the linear sieve. The key idea is to use the Möbius inversion formula to express $M(N)$ in terms of the summatory function of the Liouville function, which can be computed using the LMO algorithm. This approach has the potential to compute $M(N)$ for N up to 10^{15} or higher, provided that sufficient computational resources are available.

7.3 Spectral Analysis and the Zeros of the Zeta Function

The explicit formula for $M(x)$ in terms of the non-trivial zeros of $\zeta(s)$ provides a direct connection between the oscillatory behaviour of $M(x)$ and the distribution of the zeros. A natural extension of our work is a spectral analysis of $M(n)$ to detect signatures of the first few non-trivial zeros.

7.3.1 Fourier Analysis of $M(n)$

The Fourier transform of the sequence $M(n)$, or of its normalised version $R(n)$, may reveal peaks at frequencies corresponding to the imaginary parts γ of the non-trivial zeros. The explicit formula suggests that $M(x)$ contains oscillatory terms of the form

$$\frac{x^\rho}{\rho \zeta'(\rho)} = \frac{x^{1/2+i\gamma}}{(1/2+i\gamma) \zeta'(1/2+i\gamma)}.$$

These terms contribute oscillations with frequencies proportional to γ in the logarithmic scale. A periodogram or a Lomb–Scargle analysis of the logarithmic series $M(e^t)$ could potentially detect these frequencies, providing a numerical verification of the explicit formula.

Specifically, consider the function

$$F(t) = e^{-t/2} M(e^t), \quad t = \log x.$$

Under the Riemann hypothesis, $F(t)$ can be expressed as a sum of oscillatory terms:

$$F(t) = \sum_{\gamma} \frac{e^{i\gamma t}}{(1/2+i\gamma) \zeta'(1/2+i\gamma)} + O(e^{-\varepsilon t}).$$

The Fourier transform of $F(t)$ would therefore exhibit peaks at the frequencies γ , the imaginary parts of the non-trivial zeros. Detecting these peaks in the empirical data would constitute a direct numerical verification of the explicit formula and would provide an independent check of the Riemann hypothesis in the computed range.

7.3.2 The Gram–Schmidt and Podkletnov–Chudakov Methods

More sophisticated spectral methods, such as the Gram–Schmidt orthogonalisation or the Podkletnov–Chudakov method, have been proposed for extracting the contribution of individual zeros from the summatory function. Applying these methods to our data set could yield estimates of the first few $\zeta'(1/2+i\gamma)$ values, which are of independent interest in computational number theory.

The Gram–Schmidt method involves projecting $F(t)$ onto a basis of functions of the form $e^{i\gamma t}$, allowing the extraction of the coefficients corresponding to individual zeros. This method, while computationally intensive, has been successfully applied to other summatory functions and could provide valuable insights into the structure of the Mertens function.

7.3.3 Detecting the First Zero of $\zeta(s)$

The first non-trivial zero of $\zeta(s)$, with $\gamma_1 \approx 14.134725$, is expected to induce oscillations in $M(x)$ with a period of approximately

$$T = \frac{2\pi}{\log \gamma_1} \approx \frac{2\pi}{\log 14.134725} \approx 2.39$$

in the logarithmic scale. While this oscillation is too small to be visible in the raw data, a high-resolution Fourier analysis of $M(e^t)$ over the interval $t \in [0, \log 10^7]$ might reveal a weak spectral line at this frequency. The detection of such a line would constitute a direct numerical confirmation of the connection between $M(x)$ and the zeros of $\zeta(s)$.

To enhance the signal-to-noise ratio, we propose the following procedure. First, subtract the mean and the linear trend from the sequence $M(e^t)$. Then, apply a windowing function (e.g., a Hanning window) to reduce spectral leakage. Finally, compute the periodogram and search for peaks at frequencies corresponding to the known zeros γ_n . The amplitude of the peak at γ_1 would provide an estimate of $1/|\zeta'(1/2 + i\gamma_1)|$, which could be compared with known numerical values.

7.4 Machine Learning and Anomaly Detection

The application of modern machine learning techniques to the study of arithmetic functions is a rapidly growing area. The sequence $\mu(n)$, while appearing random, is generated by a deterministic rule. Machine learning models, particularly deep neural networks, may be capable of detecting subtle patterns in $\mu(n)$ that are invisible to classical statistical tests.

7.4.1 Sequence Prediction Models

Training a recurrent neural network (RNN) or a transformer model on the sequence $\mu(n)$ to predict future values could reveal whether there exist hidden regularities that can be exploited for prediction. If the sequence is truly random in the statistical sense, the prediction accuracy should approach the

baseline set by the empirical distribution. Any significant deviation from this baseline would indicate the presence of non-trivial structure.

We propose the following experimental setup. The sequence $\mu(n)$ for $n \leq 10^7$ is split into training and testing sets. The model is trained to predict $\mu(n+1)$ given the previous k values, where k is a hyperparameter. The prediction accuracy is measured by the cross-entropy loss or the classification accuracy. If the model achieves accuracy significantly above the baseline of $\max(p_{-1}, p_0, p_1) \approx 0.392$, this would indicate the presence of correlations that are not captured by the linear autocorrelation analysis.

7.4.2 Anomaly Detection

Unsupervised anomaly detection algorithms, such as autoencoders or isolation forests, can be applied to the sequence of pairs $(n, \mu(n), M(n))$ to identify regions of unusual behaviour. Such regions might correspond to intervals where $M(n)$ deviates from the expected random walk behaviour by an unusually large amount, potentially indicating the influence of a zero of $\zeta(s)$ or other arithmetic anomalies.

We propose the following methodology. Construct a feature vector for each index n containing n , $\mu(n)$, $\mu(n-1)$, $\mu(n-2)$, $M(n)$, $M(n)/\sqrt{n}$, and the local trend of $M(n)$ over a sliding window. Apply an isolation forest or a one-class SVM to identify outliers in this feature space. The outliers could then be examined manually to determine whether they correspond to known anomalies in the Mertens function or represent genuinely new phenomena.

7.4.3 Feature Extraction from Prime Factorisations

The value of $\mu(n)$ depends solely on the squarefree property and the parity of the number of prime factors of n . A machine learning model that is given access to the prime factorisation of n (or to features derived from it, such as the number of prime factors, the smallest prime factor, or the largest prime factor) might be able to predict $\mu(n)$ with higher accuracy than a model that only sees n itself. Such an analysis could reveal which aspects of the prime factorisation are most predictive of $\mu(n)$, providing insight into the structure of the Möbius function.

We propose the following experiment. For each n , compute the set of prime factors and derive features such as:

- $\omega(n)$, the number of distinct prime factors;
- $\Omega(n)$, the total number of prime factors counted with multiplicity;
- $p_{\min}(n)$, the smallest prime factor;

- $p_{\max}(n)$, the largest prime factor;
- $\text{rad}(n)$, the radical of n ;
- $\log n$, the logarithm of n ;
- the residue of n modulo small primes.

Train a classifier (e.g., a random forest or a gradient boosting machine) to predict $\mu(n)$ from these features. The feature importance scores would reveal which aspects of the prime factorisation are most predictive of $\mu(n)$, potentially uncovering new arithmetic relationships.

7.5 Probabilistic and Statistical Extensions

The probabilistic model for $\mu(n)$ can be extended and refined in several directions.

7.5.1 Testing Higher-Order Dependence

While our autocorrelation analysis confirmed the absence of pairwise dependence, higher-order dependencies may still exist. Tests for serial independence up to order k , such as the BDS test or the correlation integral method, can be applied to the sequence $\mu(n)$ to detect nonlinear dependencies. If such dependencies are present, they would indicate that the Möbius function is not merely a realisation of independent random variables but possesses more complex structure.

The BDS test, developed by Brock, Dechert, and Scheinkman, is a non-parametric test for the null hypothesis that a sequence is independently and identically distributed. It is based on the correlation integral, which measures the probability that two points in the sequence are close to each other in the phase space. The BDS test has been successfully applied to various time series and could provide a powerful diagnostic for the independence of $\mu(n)$.

7.5.2 The Law of the Iterated Logarithm for $M(n)$

The law of the iterated logarithm (LIL) predicts that for a random walk with step variance σ^2 ,

$$\limsup_{n \rightarrow \infty} \frac{|S_n|}{\sqrt{2\sigma^2 n \log \log n}} = 1 \quad \text{a.s.}$$

While $M(n)$ is deterministic, it is conjectured to satisfy an analogous LIL if the Riemann hypothesis is true and if the zeros of $\zeta(s)$ are sufficiently

well-behaved. Testing this conjecture numerically requires computing

$$\Lambda(N) = \max_{1 \leq n \leq N} \frac{|M(n)|}{\sqrt{2 \cdot (6/\pi^2) \cdot n \log \log n}}$$

for increasingly large N . Our data show that

$$\Lambda(10^7) \approx \frac{0.356}{\sqrt{2 \cdot 0.6079 \cdot \log \log 10^7}} \approx \frac{0.356}{\sqrt{2 \cdot 0.6079 \cdot 2.78}} \approx \frac{0.356}{1.838} \approx 0.194,$$

which is far below the LIL limit of 1. Extending this computation to larger N would reveal whether $\Lambda(N)$ grows towards unity, as predicted, or remains bounded below unity, which would indicate stronger cancellation.

7.5.3 The Distribution of $M(n)/\sqrt{n}$

The asymptotic distribution of $R(n) = M(n)/\sqrt{n}$ is not known. Under the random walk model, $R(n)$ is asymptotically normal, but the deterministic correlations in $\mu(n)$ may alter the limiting distribution. Large-scale simulations of $R(n)$ could be used to estimate the moments of the limiting distribution and to compare them with the predictions of the random walk model. Such an analysis would provide a statistical characterisation of the fluctuations of $M(n)$ in the asymptotic regime.

We propose the following study. Compute $R(n)$ for all $n \leq N$ for increasingly large N (up to 10^{12} using the Min-25 sieve). Estimate the empirical moments of $R(n)$:

$$m_k = \frac{1}{N} \sum_{n=1}^N \left(\frac{M(n)}{\sqrt{n}} \right)^k.$$

Compare the empirical moments with the theoretical moments of the normal distribution. Any significant deviation would indicate that the distribution of $R(n)$ is not normal, potentially revealing the influence of the zeta zeros on the fluctuations of $M(n)$.

7.6 Connections to Other Arithmetic Functions

The methods developed in this paper can be extended to other summatory functions of multiplicative functions.

7.6.1 The Summatory Liouville Function

The Liouville function $\lambda(n) = (-1)^{\Omega(n)}$, where $\Omega(n)$ is the total number of prime factors counted with multiplicity, satisfies $\lambda(n) = \mu(n)$ for squarefree n

but differs when n has repeated prime factors. The summatory function

$$L(x) = \sum_{n \leq x} \lambda(n)$$

is closely related to $M(x)$ through the identity

$$L(x) = \sum_{d \leq x} M(x/d^2).$$

A similar numerical investigation of $L(x)$ would provide complementary information about the distribution of prime factors and could be compared with our results for $M(x)$. In particular, the behaviour of $L(x)$ is known to be even more closely related to the Riemann hypothesis than $M(x)$: $L(x) = O(x^{1/2+\epsilon})$ is equivalent to RH, while the corresponding bound for $M(x)$ is only a consequence of RH.

7.6.2 The Summatory Jordan Totient Function

The Jordan totient function $J_k(n)$, defined as the number of k -tuples modulo n that are coprime to n , generalises Euler's totient function. It satisfies the identity

$$\sum_{d|n} J_k(d) = n^k,$$

which implies

$$J_k(n) = n^k \sum_{d|n} \frac{\mu(d)}{d^k}.$$

The summatory function of $J_k(n)$ can be studied using the same computational framework as $M(n)$, and the results could be compared with the known asymptotic expansions.

7.6.3 The Mertens Function for Dirichlet Characters

Generalising the Möbius function to Dirichlet characters χ , one can define

$$M_\chi(x) = \sum_{n \leq x} \mu(n) \chi(n).$$

The behaviour of $M_\chi(x)$ is related to the zeros of the Dirichlet L -functions and to the generalised Riemann hypothesis. A numerical study of $M_\chi(x)$ for various characters χ could reveal whether the randomness properties observed for $M(n)$ are universal across different arithmetic sequences.

For a primitive character χ modulo q , the function $M_\chi(x)$ satisfies an explicit formula analogous to that for $M(x)$, with the sum over zeros of $L(s, \chi)$. Computing $M_\chi(x)$ for several characters of small modulus would allow a comparison of the cancellation properties of different arithmetic sequences and could provide insights into the distribution of zeros of L -functions.

7.7 Summary of Future Directions

The avenues for future research outlined above are diverse and span multiple disciplines, from algorithmic number theory and high-performance computing to statistical analysis and machine learning. The most immediate priority is the extension of the computation to larger scales, which would provide crucial data on the growth rate of $\mathcal{M}(x)$ and the frequency of large deviations. Concurrently, the development of advanced algorithms, such as the Min-25 sieve and the LMO method, would enable computations at scales currently inaccessible to the linear sieve.

The spectral analysis of $M(n)$ offers a direct connection to the zeros of the zeta function and provides an opportunity to verify the explicit formula numerically. Machine learning techniques, while speculative, could uncover hidden patterns in the Möbius sequence that are invisible to classical statistical tests. Finally, the extension of the methodology to other arithmetic functions would broaden the scope of the investigation and reveal whether the randomness properties observed for $\mu(n)$ are universal or specific to this particular function.

Collectively, these future directions promise to deepen our understanding of the Mertens function, its connection to the Riemann hypothesis, and the broader structure of arithmetic functions in analytic number theory.

References

- [1] T. M. Apostol, *Introduction to Analytic Number Theory*. Springer-Verlag, New York, 1976.
- [2] E. C. Titchmarsh, *The Theory of the Riemann Zeta-function*, 2nd ed., revised by D. R. Heath-Brown, Oxford University Press, Oxford, 1986.
- [3] H. M. Edwards, *Riemann's Zeta Function*. Academic Press, New York, 1974.

- [4] G. H. Hardy and J. E. Littlewood, “Contributions to the theory of the Riemann zeta-function and the theory of the distribution of primes,” *Acta Math.*, vol. 41, pp. 119–196, 1918.
- [5] A. M. Odlyzko and H. J. J. te Riele, “Disproof of the Mertens conjecture,” *J. Reine Angew. Math.*, vol. 357, pp. 138–160, 1985.
- [6] J. Kotnik and J. van de Lune, “On the order of the Mertens function,” *Experiment. Math.*, vol. 13, no. 4, pp. 473–481, 2004.
- [7] J. Kotnik, “The Mertens function and its oscillations,” Ph.D. dissertation, Delft University of Technology, 2006.
- [8] A. E. Ingham, *The Distribution of Prime Numbers*. Cambridge University Press, Cambridge, 1932.
- [9] H. Davenport, *Multiplicative Number Theory*, 3rd ed., revised by H. L. Montgomery, Springer-Verlag, New York, 2000.
- [10] H. L. Montgomery and R. C. Vaughan, *Multiplicative Number Theory I: Classical Theory*. Cambridge University Press, Cambridge, 2007.
- [11] P. Sarnak, “Möbius randomness and dynamics,” *Notices Amer. Math. Soc.*, vol. 59, no. 4, pp. 534–539, 2012.
- [12] J. C. Lagarias, V. S. Miller, and A. M. Odlyzko, “Computing $\pi(x)$: the Meissel–Lehmer method,” *Math. Comp.*, vol. 44, no. 170, pp. 537–560, 1985.
- [13] A. Schönhage, “Schnelle Berechnung von Kettenbruchentwicklungen,” *Acta Informatica*, vol. 1, pp. 139–144, 1971.
- [14] H. Cohen, *A Course in Computational Algebraic Number Theory*. Springer-Verlag, Berlin, 1993.
- [15] W. A. Brock, W. D. Dechert, and J. A. Scheinkman, “A test for independence based on the correlation dimension,” *Econometric Reviews*, vol. 15, no. 3, pp. 197–235, 1996.
- [16] G. M. Ljung and G. E. P. Box, “On a measure of lack of fit in time series models,” *Biometrika*, vol. 65, no. 2, pp. 297–303, 1978.
- [17] J. Durbin and G. S. Watson, “Testing for serial correlation in least squares regression. I,” *Biometrika*, vol. 37, pp. 409–428, 1950.

- [18] OEIS Foundation Inc., “The On-Line Encyclopedia of Integer Sequences,” [Sequence A002321: Mertens’s function](#), 2024.
- [19] F. Mertens, “Über eine zahlentheoretische Funktion,” *Sitzungsber. Akad. Wiss. Wien Math.-Naturwiss. Kl.*, vol. 106, pp. 761–830, 1897.
- [20] H. von Koch, “Sur la distribution des nombres premiers,” *Acta Math.*, vol. 24, pp. 159–182, 1901.
- [21] E. Landau, *Handbuch der Lehre von der Verteilung der Primzahlen*, 2nd ed., Chelsea, New York, 1953.
- [22] K. Soundararajan, “Partial sums of the Möbius function,” *J. Reine Angew. Math.*, vol. 631, pp. 141–152, 2009.
- [23] G. Tenenbaum, *Introduction to Analytic and Probabilistic Number Theory*, 3rd ed., American Mathematical Society, Providence, RI, 2015.
- [24] A. Ivić, *The Riemann Zeta-Function: Theory and Applications*. Dover Publications, Mineola, NY, 2003.
- [25] H. L. Montgomery, “The pair correlation of zeros of the zeta function,” in *Proc. Sympos. Pure Math.*, vol. 24, pp. 181–193, 1973.
- [26] A. Selberg, “Contributions to the theory of the Riemann zeta-function,” *Arch. Math. Naturvid.*, vol. 48, no. 5, pp. 89–155, 1946.
- [27] J. van de Lune and E. Wattel, “On the numerical computation of the Mertens function,” *Math. Comp.*, vol. 23, pp. 409–413, 1969.